

Customer Review Helpfulness Analytics Platform



Customer Review Helpfulness Analytics Platform

Project Overview :-

This project implements a modern, cloud-native data engineering pipeline on Microsoft Azure. It ingests raw data from external sources, processes and refines it through a multi-layered medallion architecture, and delivers curated, analytics-ready data to business intelligence consumers.

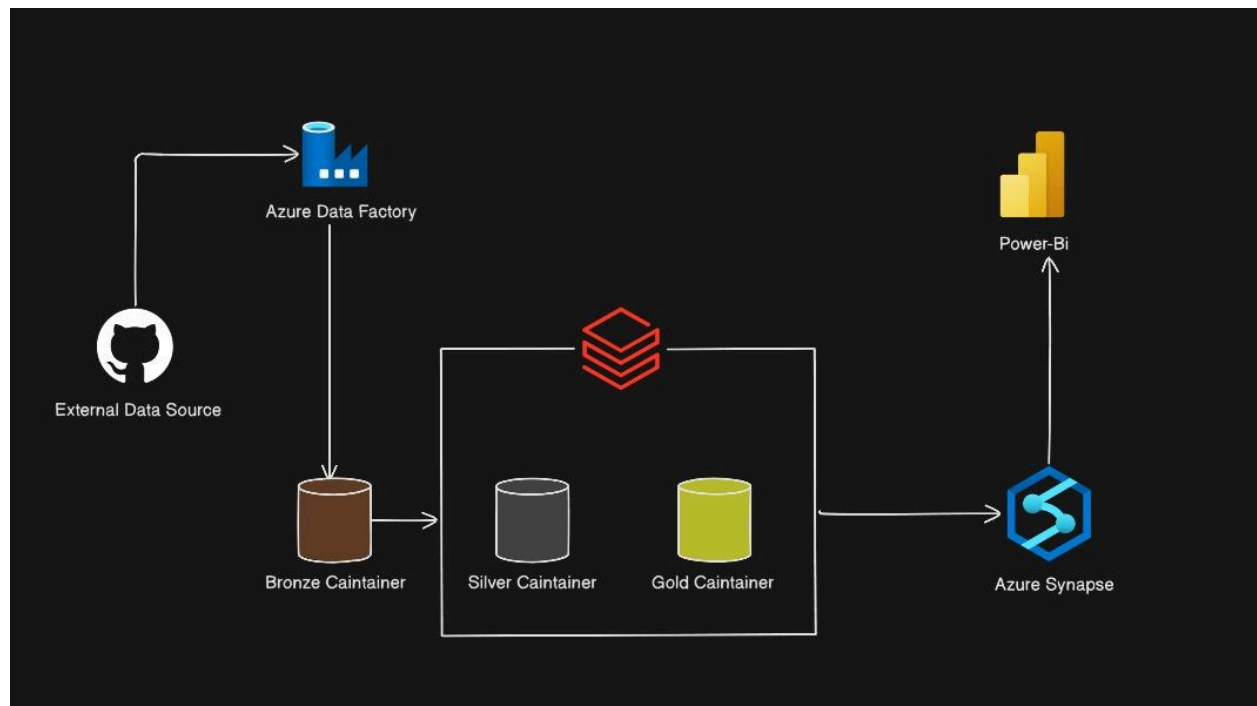
The pipeline is designed for reliability, scalability, and maintainability — enabling teams to trust their data and make faster, data-driven decisions.

Objectives :-

- Automate ingestion of raw data from external sources (e.g., GitHub repositories) into Azure using Azure Data Factory.
- Implement a Bronze → Silver → Gold medallion architecture within Azure Data Lake Storage for progressive data quality improvement.
- Apply transformation and enrichment logic using Apache Spark (Databricks) at each layer of the medallion architecture.
- Load refined, analytics-ready data into Azure Synapse Analytics for high-performance querying.
- Enable business intelligence reporting and dashboarding through Power BI integration.
- Ensure the pipeline is orchestrated, monitored, and version-controlled for operational excellence.

Architecture Overview :-

The solution follows the Medallion Architecture pattern — a widely adopted data engineering practice that organises data into three progressively refined layers (Bronze, Silver, Gold). The diagram below illustrates the end-to-end data flow.



Data Flow :-

Step 1 — Ingestion

Azure Data Factory connects to the external data source (e.g., a GitHub repository) and pulls raw data files or records into the Bronze Container in Azure Data Lake Storage.

Step 2 — Bronze Layer (Raw)

Data lands in the Bronze container in its original, unmodified form. No transformation is applied at this stage, preserving full fidelity of the source data for audit and reprocessing.

Step 3 — Silver Layer (Cleansed)

Spark jobs read from Bronze and apply data quality rules: schema validation, null handling, deduplication, and type standardisation. The cleaned output is written to the Silver container.

Step 4 — Gold Layer (Enriched)

Further Spark transformations aggregate, join, and enrich Silver data into business-domain models. These curated datasets are written to the Gold container, ready for consumption.

Step 5 — Analytics (Synapse)

Azure Synapse Analytics ingests the Gold layer data and exposes it via SQL endpoints for high-performance analytical queries.

Step 6 — Reporting (Power BI)

Power BI connects to Synapse Analytics and renders interactive dashboards and reports for business stakeholders.

Pipeline Summary :-

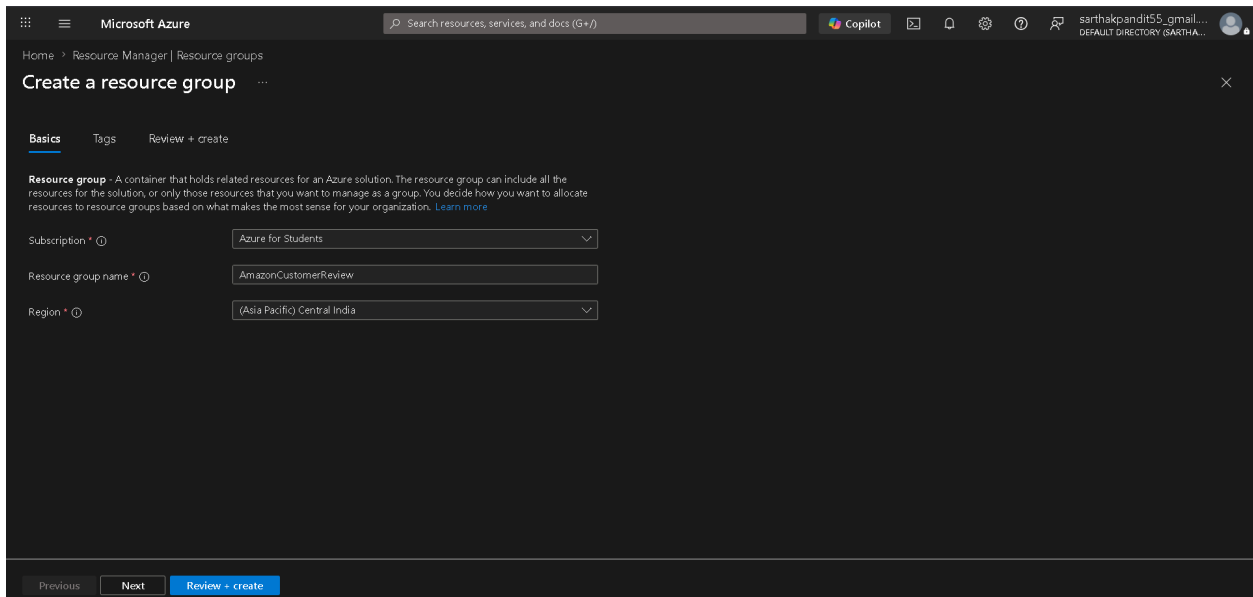
This pipeline embodies modern data engineering best practices: immutable raw storage, progressive zone-based refinement, separation of batch and streaming concerns, idempotent loads, and a single queryable reporting endpoint. The deliberate choice to combine a real open dataset (batch) with a simulated stream (Event Hubs) ensures the pipeline demonstrates both halves of a production-grade data platform — not just the easier batch-only path.

**Raw Data → Cloud Storage (ADLS Gen2) → ETL / PySpark (Databricks)
→ Database (Azure SQL) → Analytics → Dashboard (Power BI) →
Documentation**

Phase-1 (Bronze)

Task1 :- Creating a resource group

Step 1 :- Create a new Resource Group.



Microsoft Azure

Search resources, services, and docs (G+/)

Home > Resource Manager | Resource groups

Create a resource group

Basics Tags Review + create

Resource group - A container that holds related resources for an Azure solution. The resource group can include all the resources for the solution, or only those resources that you want to manage as a group. You decide how you want to allocate resources to resource groups based on what makes the most sense for your organization. [Learn more](#)

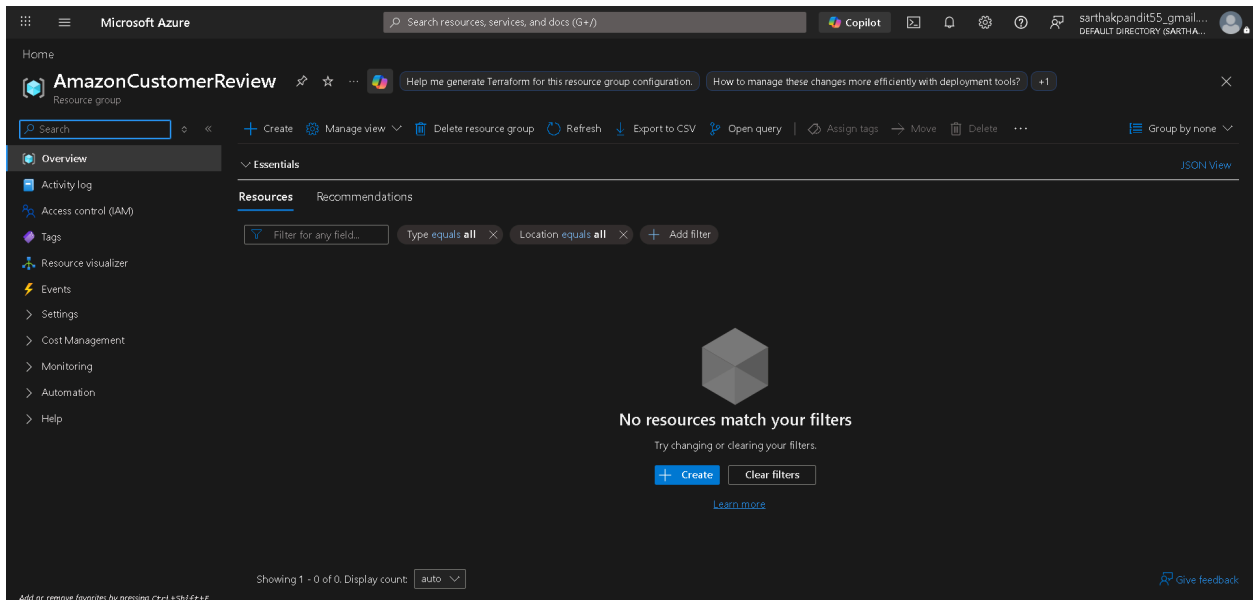
Subscription *

Resource group name *

Region *

Previous Next **Review + create**

Step 2 :- Created Resource Group named “AmazonCustomerReview”.



Microsoft Azure

Search resources, services, and docs (G+/)

Home

AmazonCustomerReview

Resource group

Help me generate Terraform for this resource group configuration. How to manage these changes more efficiently with deployment tools? +1

Search Create Manage view Delete resource group Refresh Export to CSV Open query Assign tags Move Delete Group by none

Overview

Activity log

Access control (IAM)

Tags

Resource visualizer

Events

Settings

Cost Management

Monitoring

Automation

Help

Essentials

Resources Recommendations

Filter for any field... Type equals all Location equals all Add filter

No resources match your filters

Try changing or clearing your filters.

Create Clear filters

Learn more

Showing 1 - 0 of 0. Display count: auto

Add or remove favorites by pressing **Ctrl+L+Shift+F**

Give feedback

Task2 :- Creating Storage Account (DATALAKE)

Step 1 :- Create a Storage Account “customerreviewdatalake”.

The screenshot shows the 'Create a storage account' wizard in the Microsoft Azure portal. The 'Basic' tab is selected. The form includes the following fields and options:

- Subscription:** Azure for Students
- Resource group:** AmazonCustomerReview (with a 'Create new' link)
- Storage account name:** customerreviewdatalake
- Region:** (Asia Pacific) Central India (with a 'Deploy to an Azure Extended Zone' link)
- Preferred storage type:** Azure Blob Storage or Azure Data Lake Storage (with a help icon and text: 'This helps us provide relevant guidance. It doesn't restrict your storage to this resource type. Learn more.'))
- Performance:** Standard (Recommended for most scenarios (general-purpose v2 account)). Premium (Recommended for scenarios that require low latency.) is also visible.
- Redundancy:** Locally redundant storage (LRS)

At the bottom, there are 'Previous', 'Next', and 'Review + create' buttons, along with a 'Give feedback' link.

Step 2 :- Check the requirement “**Enable Hierarchical Namespace**” to create Datalake.

The screenshot shows the 'Create a storage account' wizard in the Microsoft Azure portal, with the 'Advanced' tab selected. The form includes the following sections and options:

- Hierarchical Namespace:** A description states it is complemented by Data Lake Storage Gen2 endpoint, enables file and directory semantics, accelerates big data analytics workloads, and enables access control lists (ACLs). The checkbox 'Enable hierarchical namespace' is checked.
- Access protocols:** A note states 'Blob and Data Lake Gen2 endpoints are provisioned by default'. Below are checkboxes for 'Enable SFTP' and 'Enable network file system v3', both of which are unchecked.
- Blob storage:** A checkbox for 'Allow cross-tenant replication' is unchecked. A message below states: 'Cross-tenant replication and hierarchical namespace cannot be enabled simultaneously.'
- Access tier:** Three radio buttons are present: 'Hot' (selected, 'Optimized for frequently accessed data and everyday usage scenarios'), 'Cool' ('Optimized for infrequently accessed data and backup scenarios'), and 'Cold' ('Optimized for rarely accessed data and backup scenarios').
- Azure Files:** A checkbox for 'Enable Managed Identity for SMB' is unchecked.
- Require Encryption in Transit for SMB:** This checkbox is checked.

At the bottom, there are 'Previous', 'Next', and 'Review + create' buttons, along with a 'Give feedback' link.

Step 3 :- Check **Create**.

The screenshot shows the 'Create a storage account' wizard in the Microsoft Azure portal. The 'Review + create' tab is selected, displaying a summary of the configuration. The 'Basics' section includes: Subscription (Azure for Students), Resource group (AmazonCustomerReview), Location (Central India), Storage account name (customerreviewdatalake), Primary service (Azure Blob Storage or Azure Data Lake Storage), Performance (Standard), and Replication (Locally redundant storage (LRS)). The 'Advanced' section shows various settings like 'Enable hierarchical namespace' (Enabled), 'Enable SFTP' (Disabled), 'Enable network file system v3' (Disabled), 'Allow cross-tenant replication' (Disabled), 'Access tier' (Hot), and 'Managed identity for SMB' (Disabled). At the bottom, there are 'Previous', 'Next', and 'Create' buttons, along with a 'Give feedback' link.

Section	Setting	Value
Basics	Subscription	Azure for Students
	Resource group	AmazonCustomerReview
	Location	Central India
	Storage account name	customerreviewdatalake
	Primary service	Azure Blob Storage or Azure Data Lake Storage
	Performance	Standard
	Replication	Locally redundant storage (LRS)
Advanced	Enable hierarchical namespace	Enabled
	Enable SFTP	Disabled
	Enable network file system v3	Disabled
	Allow cross-tenant replication	Disabled
	Access tier	Hot
	Managed identity for SMB	Disabled

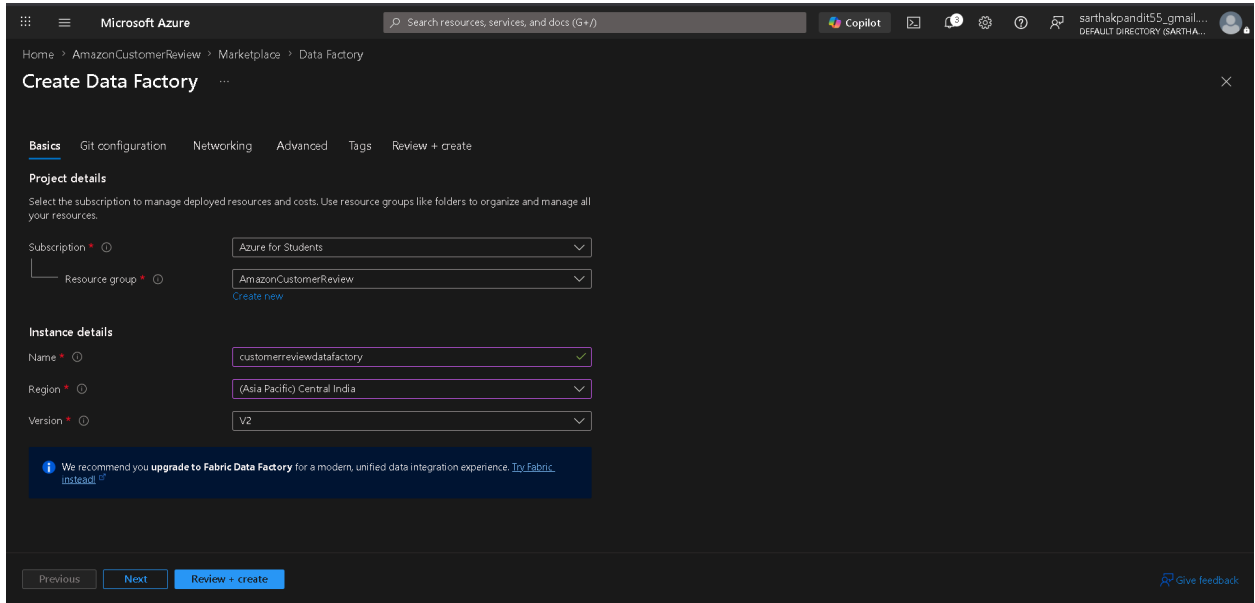
Step 4 :- Created Storage Account (DataLake) named “customerreviewdatalake”.

The screenshot shows the overview page for the 'customerreviewdatalake' storage account in the Microsoft Azure portal. The 'Overview' tab is selected, displaying a summary of the account's properties. The 'Essentials' section includes: Resource group (AmazonCustomerReview), Location (centralindia), Subscription (Azure for Students), Subscription ID (970f8371-0e8a-4065-8242-910c022190ac), Disk state (Available), and Tags (Add tags). The 'Properties' section shows the account's configuration: Performance (Standard), Replication (Locally redundant storage (LRS)), Account kind (StorageV2 (general purpose v2)), Provisioning state (Succeeded), and Created (24/05/2026, 15:15:19). The 'Data Lake Storage' section lists various settings like 'Hierarchical namespace' (Enabled), 'Default access tier' (Hot), 'Blob anonymous access' (Disabled), 'Blob soft delete' (Enabled (7 days)), 'Container soft delete' (Enabled (7 days)), 'Versioning' (Disabled), 'Change feed' (Disabled), 'NFS v3' (Disabled), and 'SFTP' (Disabled). The 'Security' section shows 'Require secure transfer for REST API operations' (Enabled), 'Storage account key access' (Enabled), 'Minimum TLS version' (Version 1.2), and 'Infrastructure encryption' (Disabled). The 'Networking' section shows 'Public network access' (Enabled), 'Public network access scope' (Enable from all networks), and 'Private endpoint connections' (n). At the bottom, there is a note: 'Add or remove favorites by pressing Ctrl+Shift+F'.

Section	Property	Value
Essentials	Resource group	AmazonCustomerReview
	Location	centralindia
	Subscription	Azure for Students
	Subscription ID	970f8371-0e8a-4065-8242-910c022190ac
	Disk state	Available
	Tags	Add tags
Properties	Performance	Standard
	Replication	Locally redundant storage (LRS)
Data Lake Storage	Account kind	StorageV2 (general purpose v2)
	Provisioning state	Succeeded
Security	Created	24/05/2026, 15:15:19

Task3 :- Creating Azure Data Factory

Step 1 :- Create Azure Data Factory “customerreviewdatafactory”.



Microsoft Azure

Search resources, services, and docs (G+/)

Home > AmazonCustomerReview > Marketplace > Data Factory

Create Data Factory

Basics Git configuration Networking Advanced Tags Review + create

Project details

Select the subscription to manage deployed resources and costs. Use resource groups like folders to organize and manage all your resources.

Subscription: Azure for Students

Resource group: AmazonCustomerReview

Instance details

Name: customerreviewdatafactory

Region: (Asia Pacific) Central India

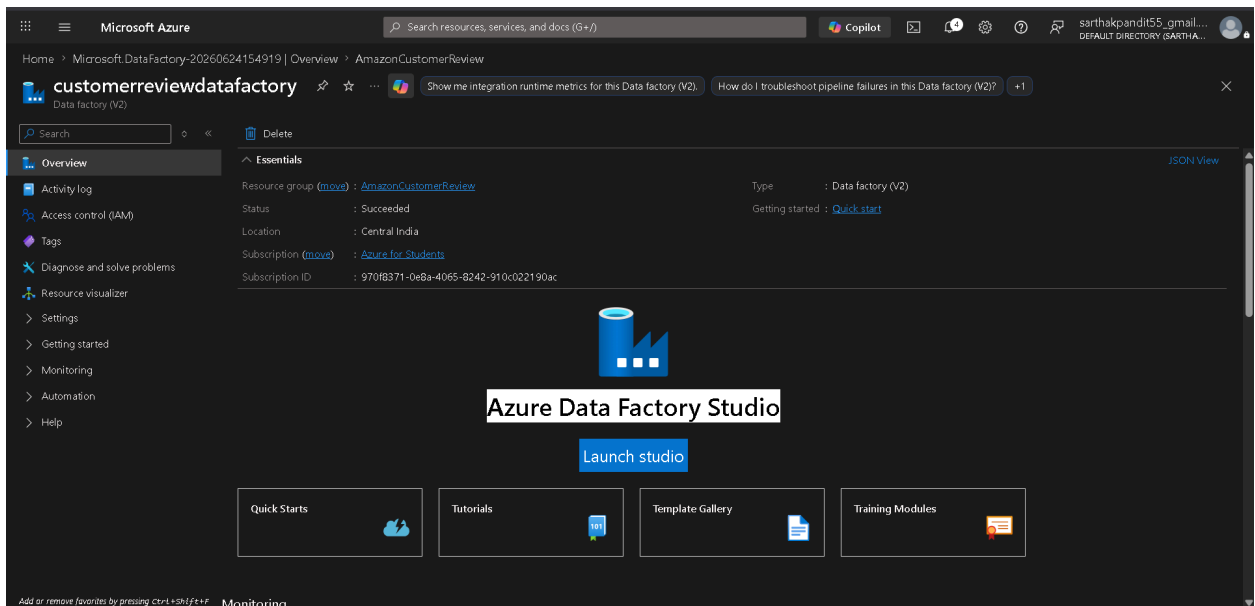
Version: V2

We recommend you **upgrade to Fabric Data Factory** for a modern, unified data integration experience. [Try Fabric instead](#)

Previous Next Review + create

Give feedback

Step 2 :- Created Azure Data Factory named “customerreviewdatafactory”.



Microsoft Azure

Search resources, services, and docs (G+/)

Home > Microsoft.DataFactory-20260624154919 | Overview > AmazonCustomerReview

customerreviewdatafactory

Data factory (V2)

Search

Delete

Overview

- Activity log
- Access control (IAM)
- Tags
- Diagnose and solve problems
- Resource visualizer
- Settings
- Getting started
- Monitoring
- Automation
- Help

Essentials

Resource group (move): AmazonCustomerReview

Status: Succeeded

Location: Central India

Subscription (move): Azure for Students

Subscription ID: 970f8371-0e8a-4065-8242-910c022190ac

Type: Data factory (V2)

Getting started: [Quick start](#)

Azure Data Factory Studio

Launch studio

Quick Starts

Tutorials

Template Gallery

Training Modules

Add or remove favorites by pressing **Ctrl+Shift+F**

Monitoring

Task4:- Creating Containers in DataLake

Step 1 :- Create a container named “**bronze**”.

Microsoft Azure

Search resources, services, and docs (G+/)

Home > AmazonCustomerReview > Marketplace > Data Factory

Create Data Factory

Basics | Git configuration | Networking | Advanced | Tags | Review + create

Project details

Select the subscription to manage deployed resources and costs. Use resource groups like folders to organize and manage all your resources.

Subscription *

Resource group * [Create new](#)

Instance details

Name *

Region *

Version *

Recommendation: We recommend you **upgrade to Fabric Data Factory** for a modern, unified data integration experience. [Try Fabric instead](#)

[Previous](#) [Next](#) [Review + create](#) [Give feedback](#)

Step 2 :- Create a container named “**silver**”.

Microsoft Azure

Search resources, services, and docs (G+/)

Home > AmazonCustomerReview > customerreviewdatalake

customerreviewdatalake | Containers

Storage account

Search

+ Add container | Upload | Refresh | Delete | Change access level | Restore containers | Edit columns

Showing all 2 items

☐	Name	Last modified	Anonymous access level
☐	logs	24/06/2026, 15:15:44	Private
☐	bronze	24/06/2026, 16:09:28	Private

New container

Name *

Anonymous access level *

Message: The access level is set to private because anonymous access is disabled on this storage account.

Advanced

[Create](#) [Give feedback](#)

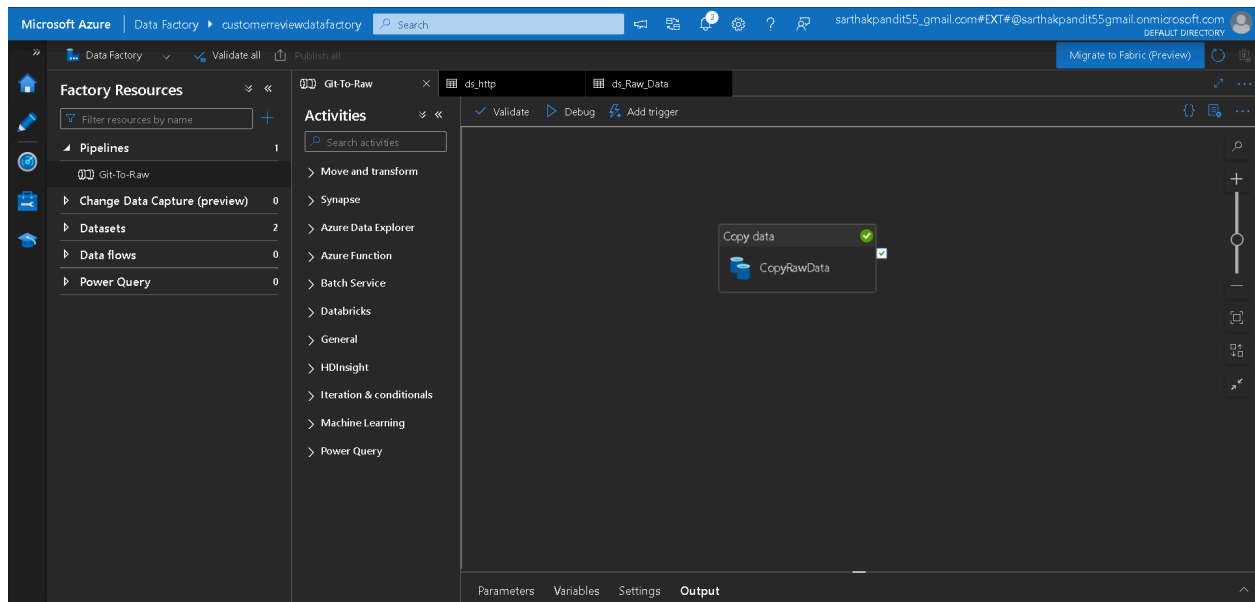
Step 3 :- Create a container named “gold”.

The screenshot shows the Microsoft Azure portal interface. The left sidebar contains navigation options: Overview, Activity log, Tags, Diagnose and solve problems, Access Control (IAM), Data migration, Events, Storage browser, Partner solutions, Resource visualizer, Data storage, Containers (selected), Classic file shares, Queues, Tables, Security + networking, and Data management. The main area displays the 'customerreviewdatalake | Containers' page. A table lists existing containers: 'logs', 'bronze', and 'silver'. The 'New container' dialog box is open on the right, showing the 'Name' field set to 'gold' and the 'Anonymous access level' set to 'Private (no anonymous access)'. A message indicates that anonymous access is disabled on this storage account. The 'Create' button is at the bottom of the dialog.

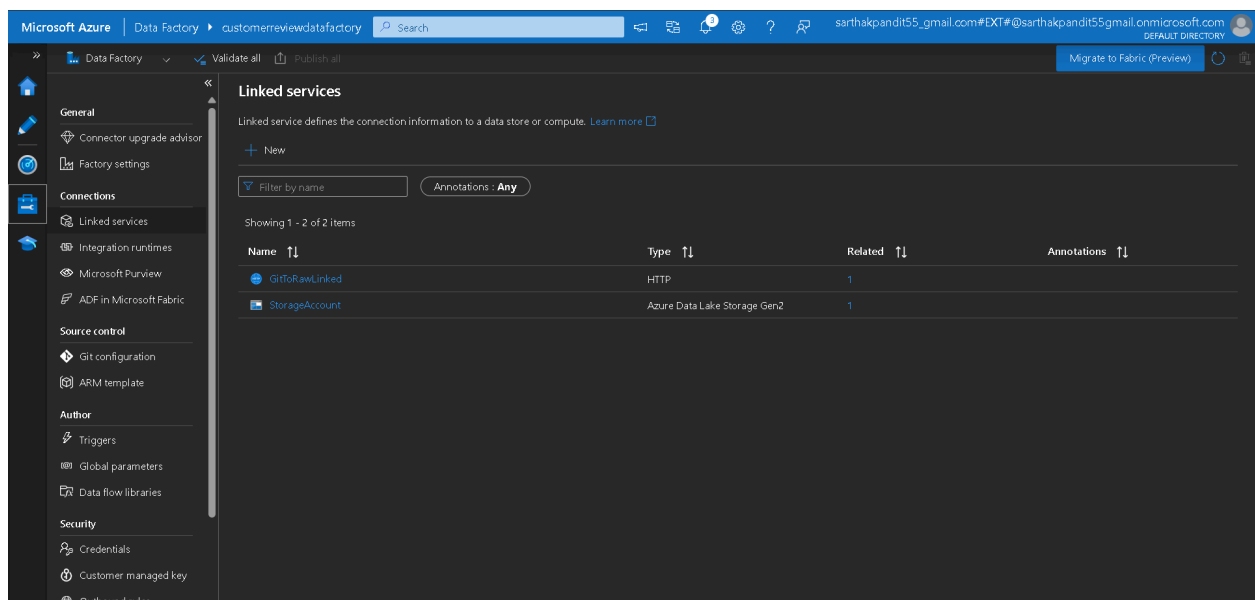
Name	Last modified	Anonymous access level
logs	24/06/2026, 15:15:44	Private
bronze	24/06/2026, 16:09:28	Private
silver	24/06/2026, 16:09:47	Private

Task5 :- Get Data from Source

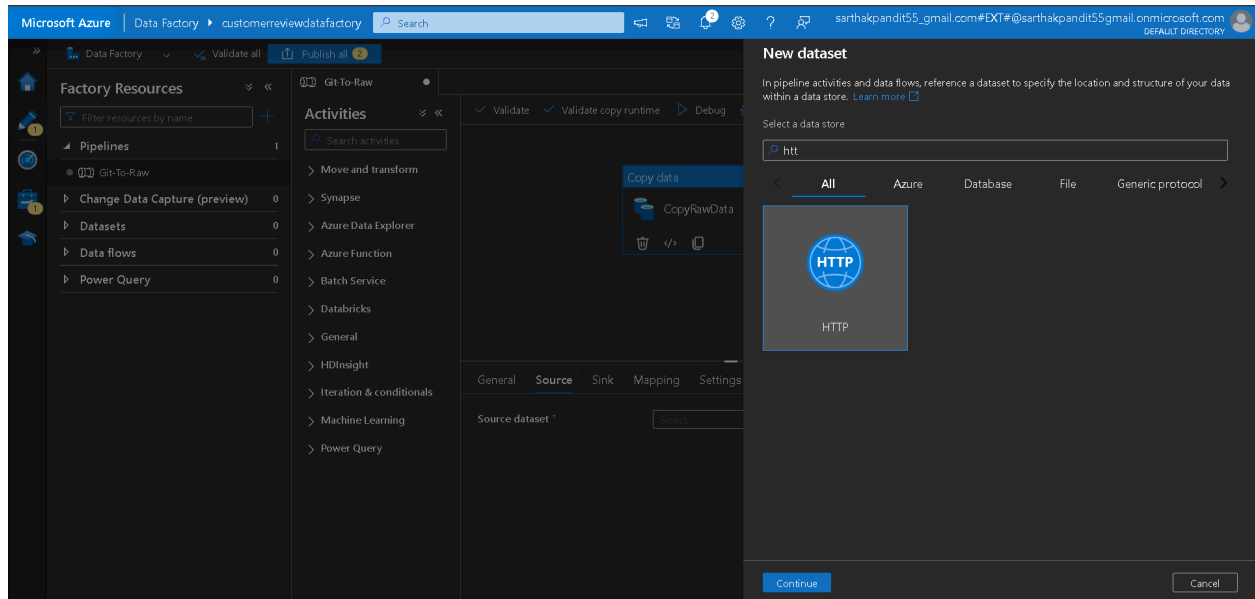
Step 1 :- Create a pipeline named “**Git-To-Raw**” and select “**Copy Activity**”.



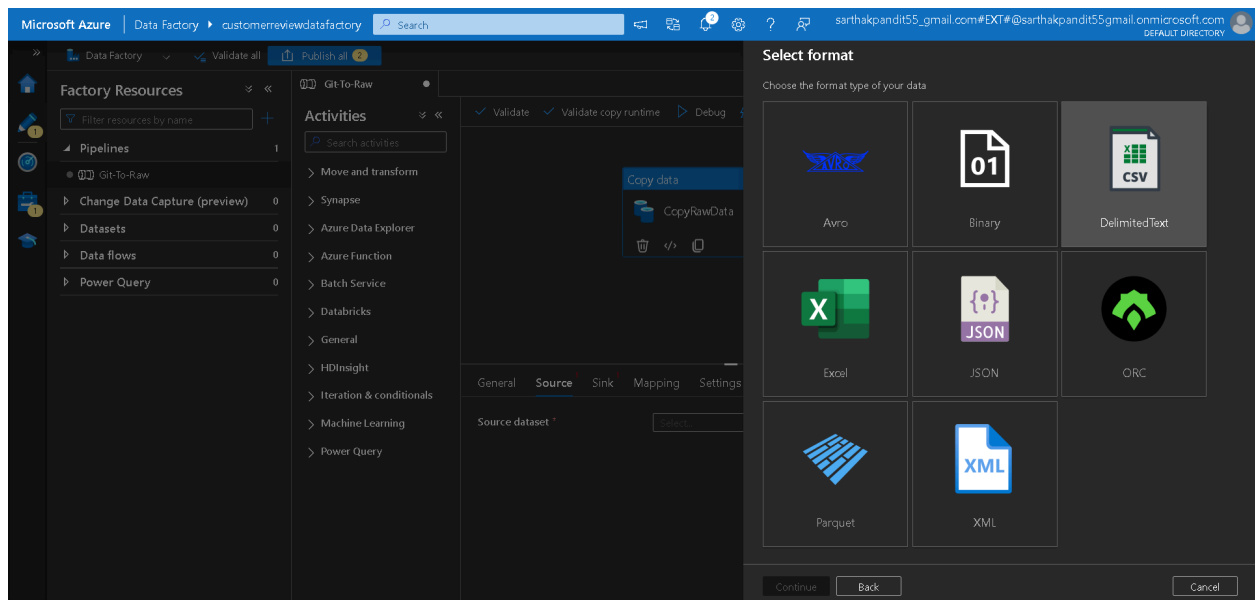
Step 2 :- Go to the “**Manage**” tab, Create linked services named “**GitToRawLinked**” to connect with source and “**StorageAccount**” to connect with destination.



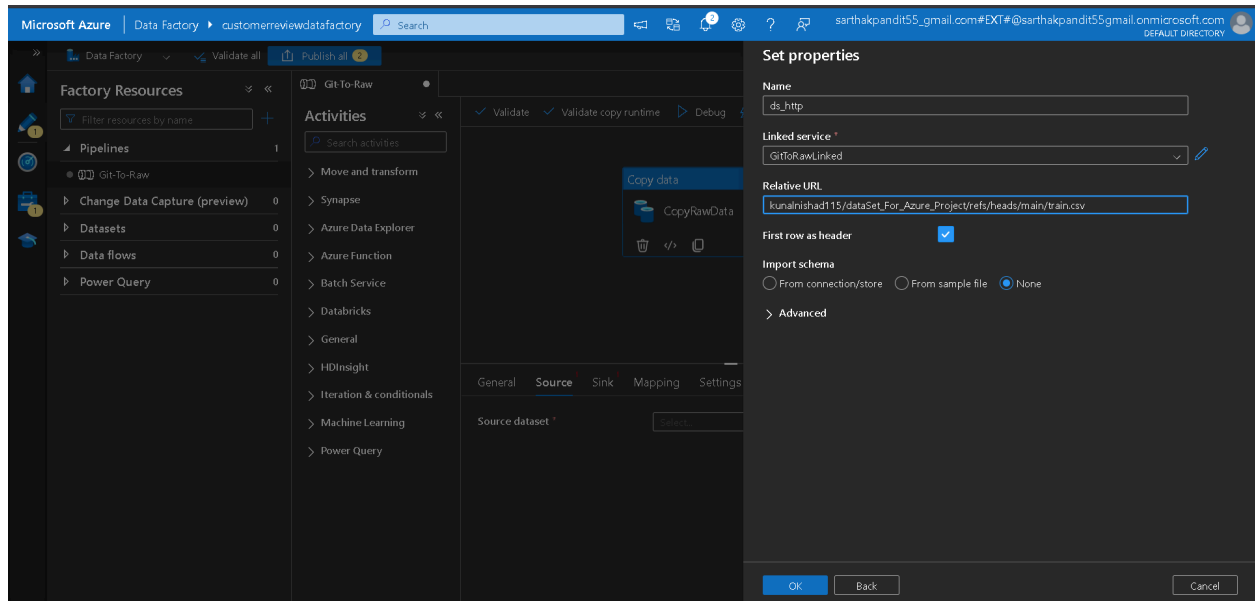
Step 3 :- Select the Data Store “**HTTP**” for Source.



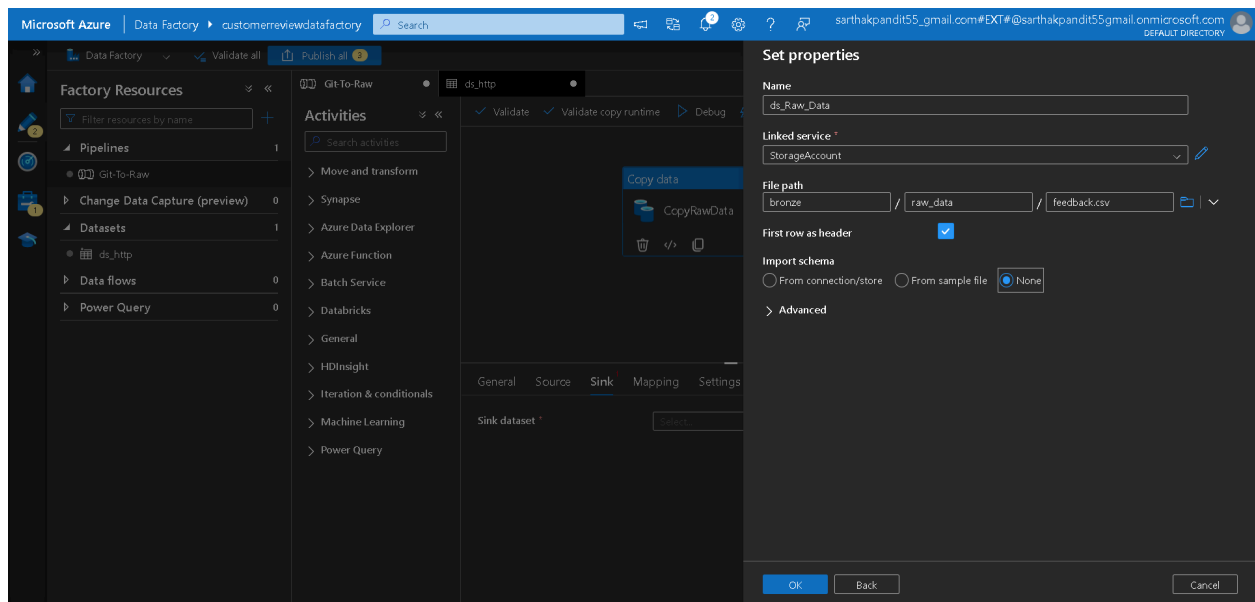
Step 4 :- Select the data format type “**CSV**”.



Step 5 :- Provide “Related URL”.



Step 6 :- Select the location to store “Raw Data” .



Step 7 :- Click “**Debug**” to run the pipeline.

The screenshot shows the Microsoft Azure Data Factory interface. On the left, the 'Factory Resources' pane lists 'Pipelines' (1), 'Datasets' (2), 'Data flows' (0), and 'Power Query' (0). The 'Activities' pane shows a search bar and a list of activity categories: Move and transform, Synapse, Azure Data Explorer, Azure Function, Batch Service, Databricks, General, HDInsight, Iteration & conditionals, Machine Learning, and Power Query. The main canvas displays a 'Copy data' activity connected to a 'CopyRawData' dataset. Below the canvas, the 'Output' tab shows the pipeline run ID '280ba7eb-ea9c-45ed-b33e-7ad6fc795146' and a status of 'Succeeded'. A table below shows the activity details:

Activity name	Activity st...	Activit...	Run start	Duration	Integration runtime
CopyRawData	Succeeded	Copy data	6/24/2026, 4:55:29 PM	15s	AutoResolveIntegrationRuntime (Central Indi

Step 8 :- Extraction of Data is complete and stored in a “**bronze**” container.

The screenshot shows the Microsoft Azure portal interface. The left sidebar shows the navigation pane with 'Overview', 'Diagnose and solve problems', 'Access Control (IAM)', and 'Settings'. The main area displays the 'bronze' container. The 'Overview' tab is selected, showing a search bar, a list of actions (Add Directory, Upload, Refresh, Delete, Copy, Paste, Rename, Acquire lease, Break lease, Edit columns), and a table of blobs. The table shows one item:

Name	Last modified	Access tier	Blob type	Size	Lease state
feedback.csv	24/06/2026, 16:55:41	Hot (Inferred)	Block blob	3.92 MiB	Available

Phase-2 (Silver)

Task1 :- Create Azure Databricks

Step 1 :- Create Azure Databricks “CustomerReviewDatabricks”.

Microsoft Azure

Home > AmazonCustomerReview > Marketplace > Azure Databricks

Create an Azure Databricks workspace

Basics Networking Encryption Security & compliance Tags Review + create

Project Details

Select the subscription to manage deployed resources and costs. Use resource groups like folders to organize and manage all your resources.

Subscription: Azure for Students

Resource group: AmazonCustomerReview

Instance Details

Workspace name: CustomerReviewDatabricks

Region: Central India

Pricing Tier: Trial (Premium - 14-Days Free DBU)

Workspace type

☐ Serverless
No setup required. Uses Databricks-managed default storage and serverless compute. Connect to your cloud storage at any time. [Learn more](#)

☒ Hybrid
Requires access to compute and storage in your Azure account. Supports custom compute.

Managed Resource Group name: managed-adb-customerreview

[Review + create](#) < Previous Next > Networking

Step 2 :- Click “Create” to create Azure Databricks resource.

Microsoft Azure

Home > AmazonCustomerReview > Marketplace > Azure Databricks

Create an Azure Databricks workspace

Basics Networking Encryption Security & compliance Tags [Review + create](#)

Validation Succeeded

Summary

Basics

Workspace name	CustomerReviewDatabricks
Subscription	Azure for Students
Resource group	AmazonCustomerReview
Region	Central India
Pricing Tier	trial
Workspace type	Hybrid
Managed Resource Group name	managed-adb-customerreview

Networking

Deploy Azure Databricks workspace with Secure Cluster Connectivity (No Public IP)	No
Deploy Azure Databricks workspace in your own Virtual Network (VNet)	No

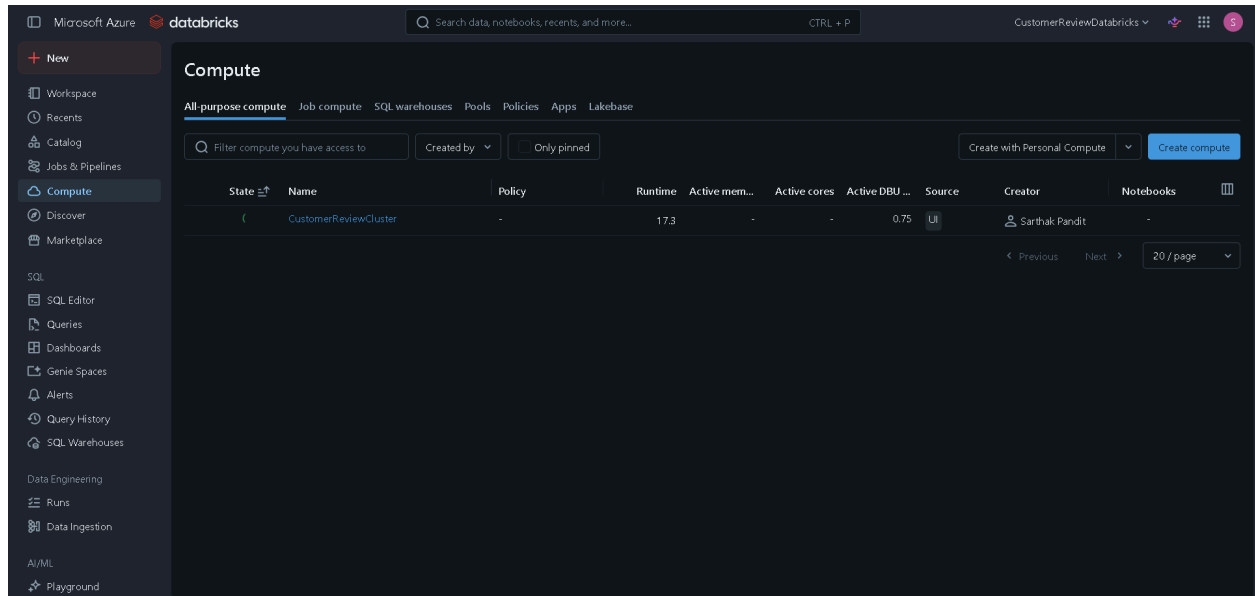
[Create](#) < Previous Download a template for automation

Step 3 :- Created Azure Databricks named “AmazonCustomerReview” .

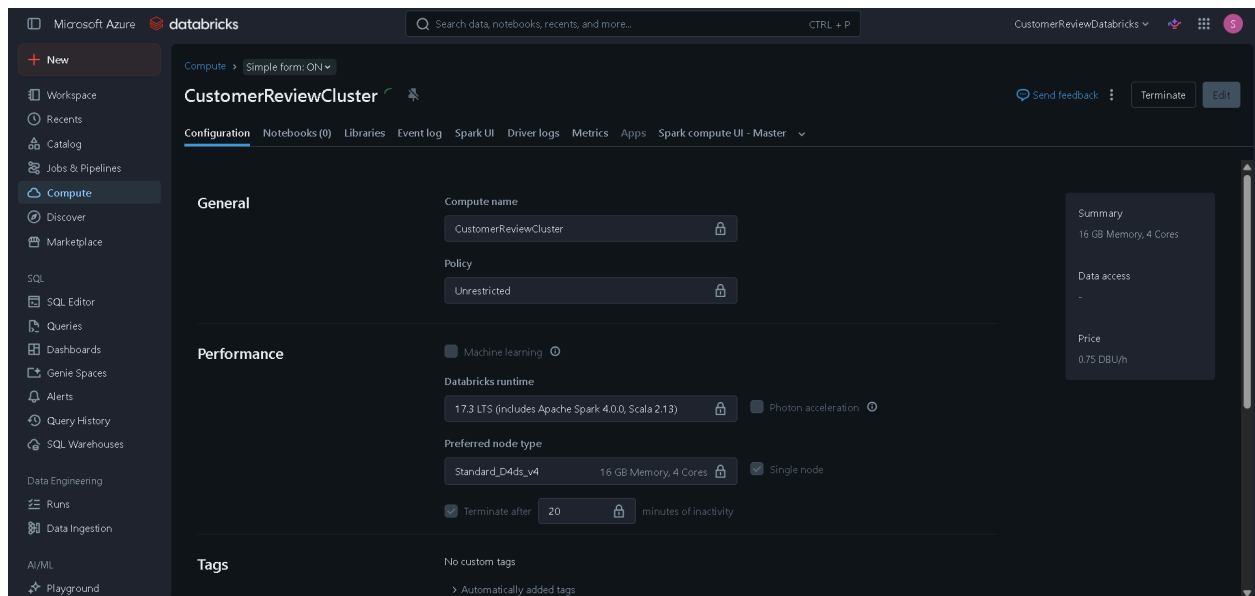
The screenshot displays the Microsoft Azure portal interface for the 'CustomerReviewDatabricks' resource. The top navigation bar includes the Microsoft Azure logo, a search bar, and user information. The main content area is divided into a left sidebar with navigation options (Overview, Activity log, Access control (IAM), Tags, Diagnose and solve problems, Resource visualizer, Settings, Automation, Help) and a central pane. The central pane shows the 'Overview' tab for the 'CustomerReviewDatabricks' resource, which is an 'Azure Databricks Service'. The 'Essentials' section lists key properties: Status (Active), Resource group (AmazonCustomerReview), Location (Central India), Subscription (Azure for Students), and Subscription ID (970f8371-0e8a-4065-8242-910c022190ac). It also displays Workspace type (Hybrid), Managed Resource Group (managed-adb-customerreview), URL (https://adb-7405615961048449.9.azuredatabricks.net), and Pricing Tier (Trial (Premium - 14-Days Free DBUs) (Click to change)). Below the essentials, there is a large red Databricks logo and two buttons: 'Launch Workspace' and 'Upgrade to Premium'. At the bottom, there are four action buttons: 'Documentation', 'Getting Started', 'Import Data from File', and 'Import Data from Azure Storage'. The bottom status bar indicates 'Add or remove favorites by pressing Ctrl+Shift+F'.

Task2 :- Create new Cluster in Azure Databrick

Step 1 :- Go to compute, Click “**Create Cluster**”.

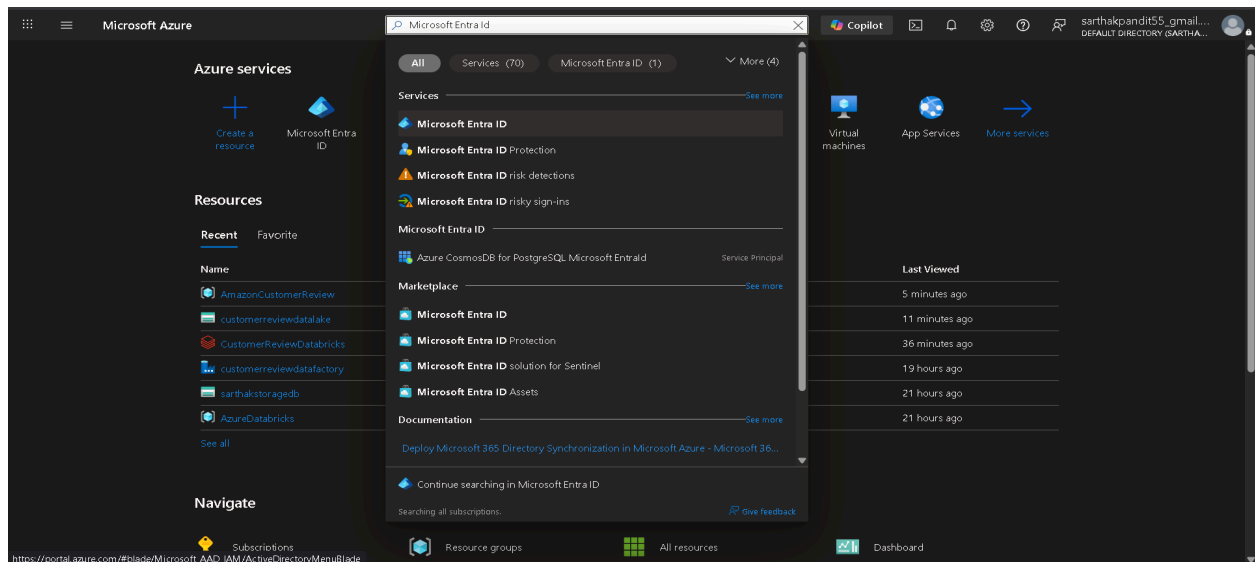


Step 2 :- Create a cluster named “**CustomerReviewCluster**”.

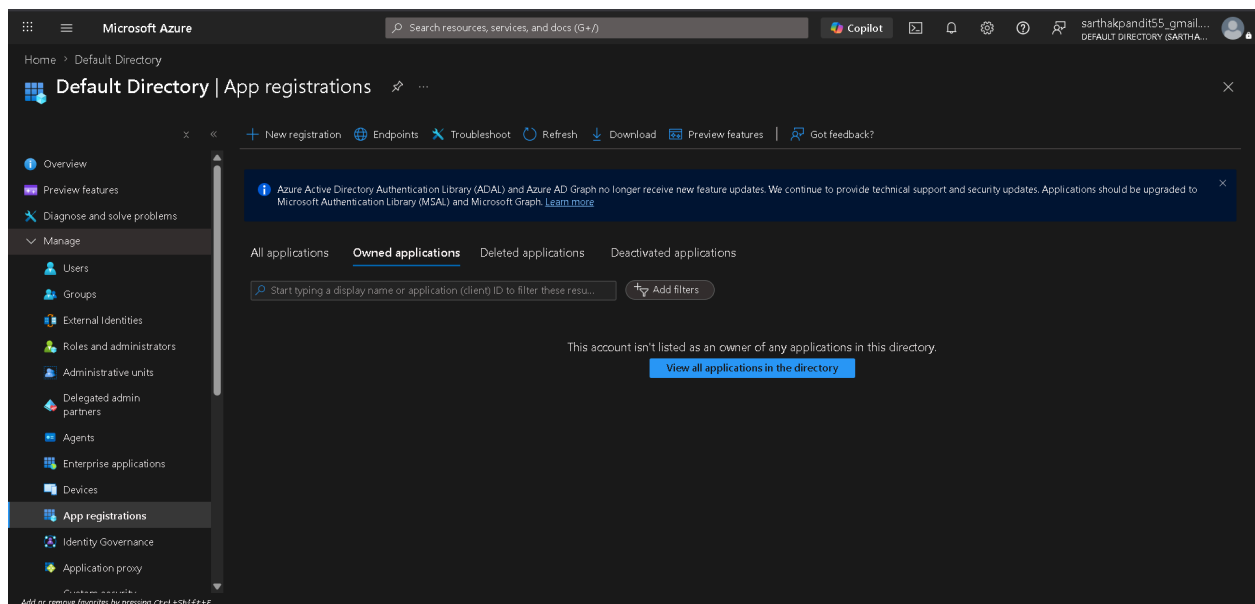


Task3 :- Providing Direct Access of Datalake to Databricks

Step 1 :- Search and Open “**Microsoft Entra ID**”.



Step 2 :- Go to the “**Manager**” tab then go to “**App Registrations**” and click “**New Registration**”.



Step 3 :- Fill the details and click ‘Register’.

The screenshot shows the 'Register an application' page in the Microsoft Azure portal. The page is titled 'Register an application' and has a close button (X) in the top right corner. The main content area is divided into sections: 'Name', 'Supported account types', and 'Redirect URI (optional)'. The 'Name' section has a text input field with 'CustomerReview-App' and a checkmark icon. The 'Supported account types' section has a dropdown menu set to 'Single tenant only - Default Directory' and a 'Help me choose' link. The 'Redirect URI (optional)' section has a 'Select a platform' dropdown and a text input field with 'e.g. https://example.com/auth'. Below these sections is a note: 'Register an app you're working on here. Integrate gallery apps and other apps from outside your organization by adding from Enterprise applications.' At the bottom, there is a 'By proceeding, you agree to the Microsoft Platform Policies' link and a blue 'Register' button.

Microsoft Azure

Search resources, services, and docs (G+/)

Copilot

Home > Default Directory | App registrations

Register an application

Name
The user-facing display name for this application (this can be changed later).

CustomerReview-App

Supported account types
Choose the account types that can use this application or access this API

Single tenant only - Default Directory [Help me choose](#)

Redirect URI (optional)
We'll return the authentication response to this URI after successfully authenticating the user. Providing this now is optional and it can be changed later, but a value is required for most authentication scenarios.

Select a platform

Register an app you're working on here. Integrate gallery apps and other apps from outside your organization by adding from [Enterprise applications](#).

By proceeding, you agree to the [Microsoft Platform Policies](#)

[Register](#)

Step 4 :- App is created and now copy the ‘Application (client) ID’ and ‘Directory ID’.

The screenshot shows the 'CustomerReview-App' overview page in the Microsoft Azure portal. The page is titled 'CustomerReview-App' and has a close button (X) in the top right corner. The left sidebar contains a search bar and a list of navigation items: Overview, Quickstart, Integration assistant, Diagnose and solve problems, Manage, and Support + Troubleshooting. The main content area is divided into sections: 'Essentials' and 'Get Started'. The 'Essentials' section displays key information about the application: Display name (CustomerReview-App), Application (client) ID (439c4141-142c-472d-8525-ab443f9b4d29), Object ID (523e9312-78a2-40ee-adb5-9a4f2c33e0f8), Directory (tenant) ID (53a18eb4-fd3a-4238-a082-0767b365db8b), Supported account types (b2b or organization only), Client credentials (Add a certificate or secret), Redirect URIs (Add a Redirect URI), Application ID URI (Add an Application ID URI), Managed application in Io... (CustomerReview-App), and State (Activated). The 'Get Started' section has links for 'Get Started' and 'Documentation'. At the bottom, there is a section titled 'Build your application with the Microsoft identity platform' with a brief description and a 'Learn more' link.

Microsoft Azure

Search resources, services, and docs (G+/)

Copilot

Home > Default Directory | App registrations

CustomerReview-App

Deactivate Delete Endpoints Preview features

Overview

Quickstart

Integration assistant

Diagnose and solve problems

Manage

Support + Troubleshooting

Essentials

Display name : CustomerReview-App

Application (client) ID : 439c4141-142c-472d-8525-ab443f9b4d29

Object ID : 523e9312-78a2-40ee-adb5-9a4f2c33e0f8

Directory (tenant) ID : 53a18eb4-fd3a-4238-a082-0767b365db8b

Supported account types : [b2b or organization only](#)

Client credentials : [Add a certificate or secret](#)

Redirect URIs : [Add a Redirect URI](#)

Application ID URI : [Add an Application ID URI](#)

Managed application in Io... : [CustomerReview-App](#)

State : [Activated](#)

Welcome to the new and improved App registrations. Looking to learn how it's changed from App registrations (Legacy)? [Learn more](#)

Azure Active Directory Authentication Library (ADAL) and Azure AD Graph no longer receive new feature updates. We continue to provide technical support and security updates. Applications should be upgraded to Microsoft Authentication Library (MSAL) and Microsoft Graph. [Learn more](#)

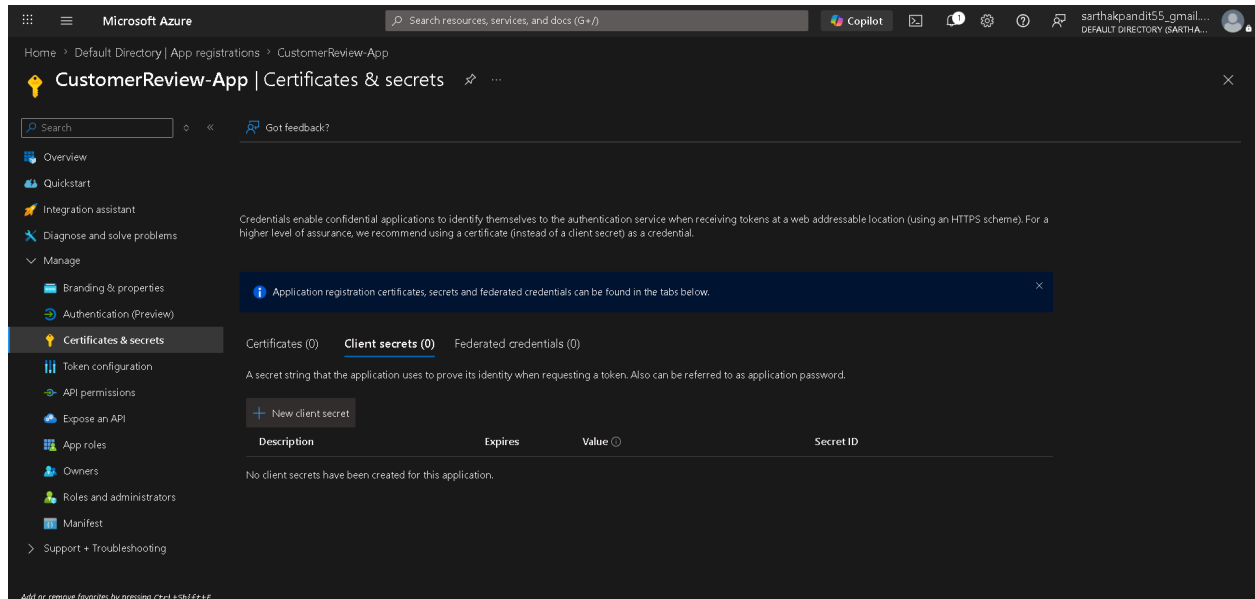
[Get Started](#) [Documentation](#)

Build your application with the Microsoft identity platform

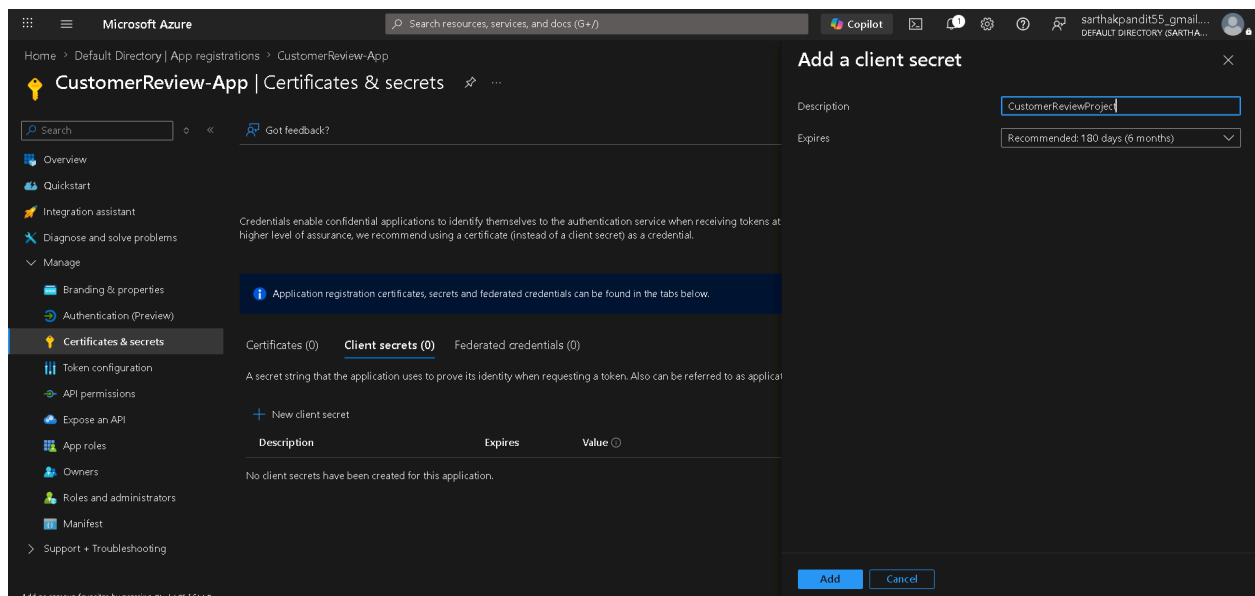
The Microsoft identity platform is an authentication service, open-source libraries, and application management tools. You can create modern, standards-based authentication solutions, access and protect APIs, and add sign-in for your users and customers. [Learn more](#)

Add or remove favorites by pressing **Ctrl+Shift+F**

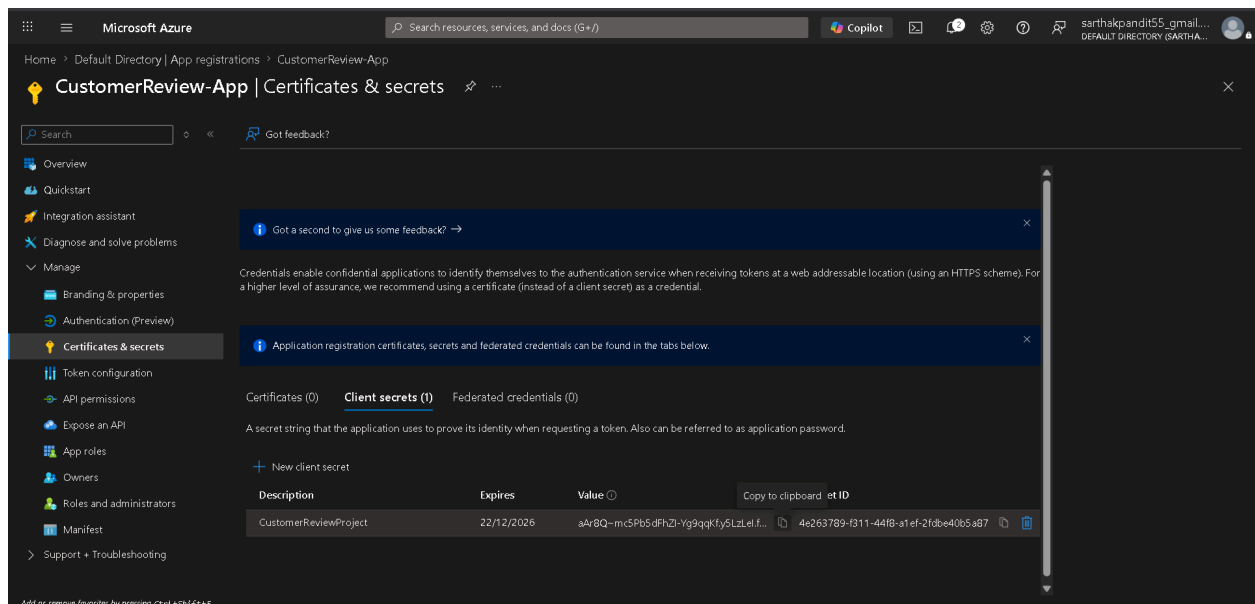
Step 5 :- Go to the “**Manager**” tab and then go to “**Certificates & Secrets**” and click “**New client secret**”.



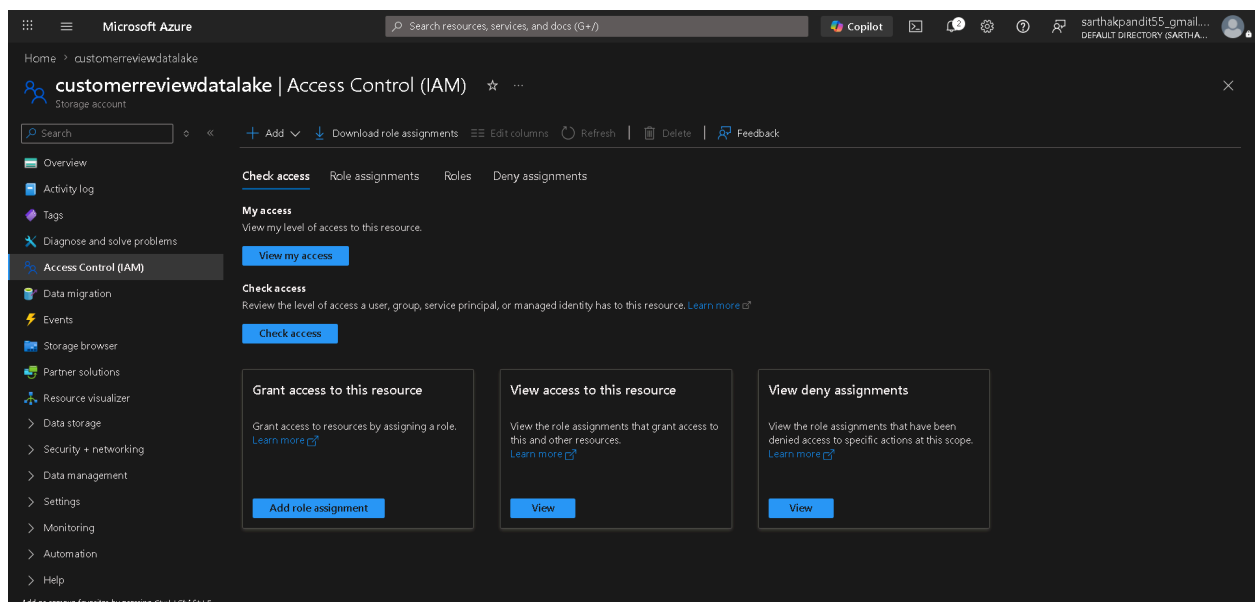
Step 6 :- Add the client secret.



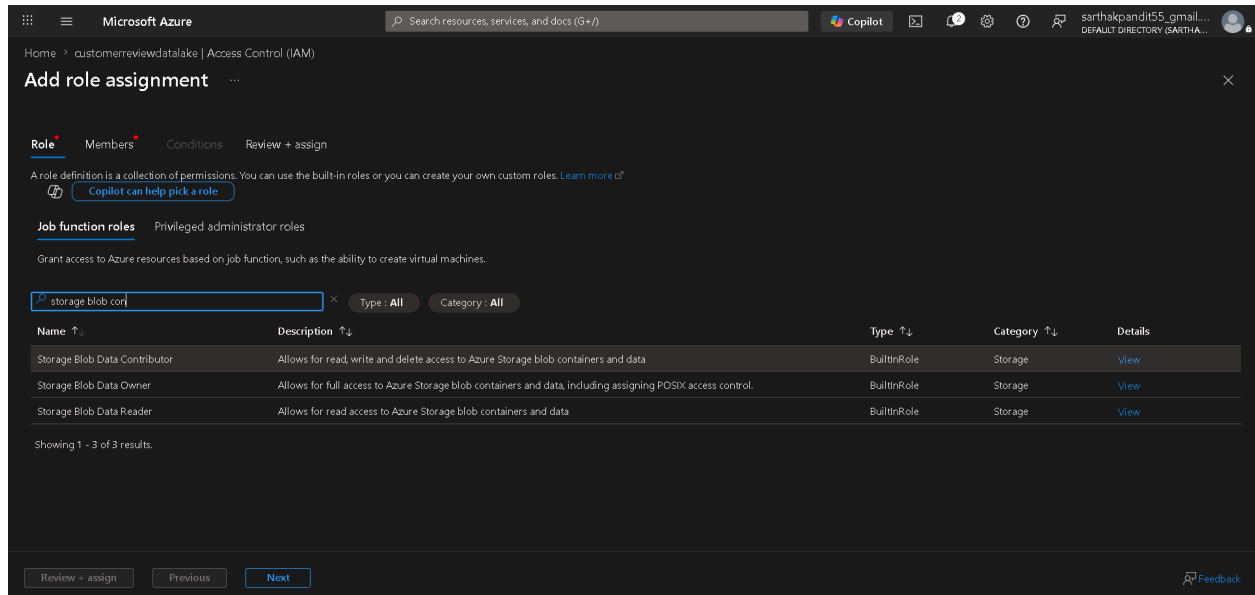
Step 7 :- Copy the value of the secret.



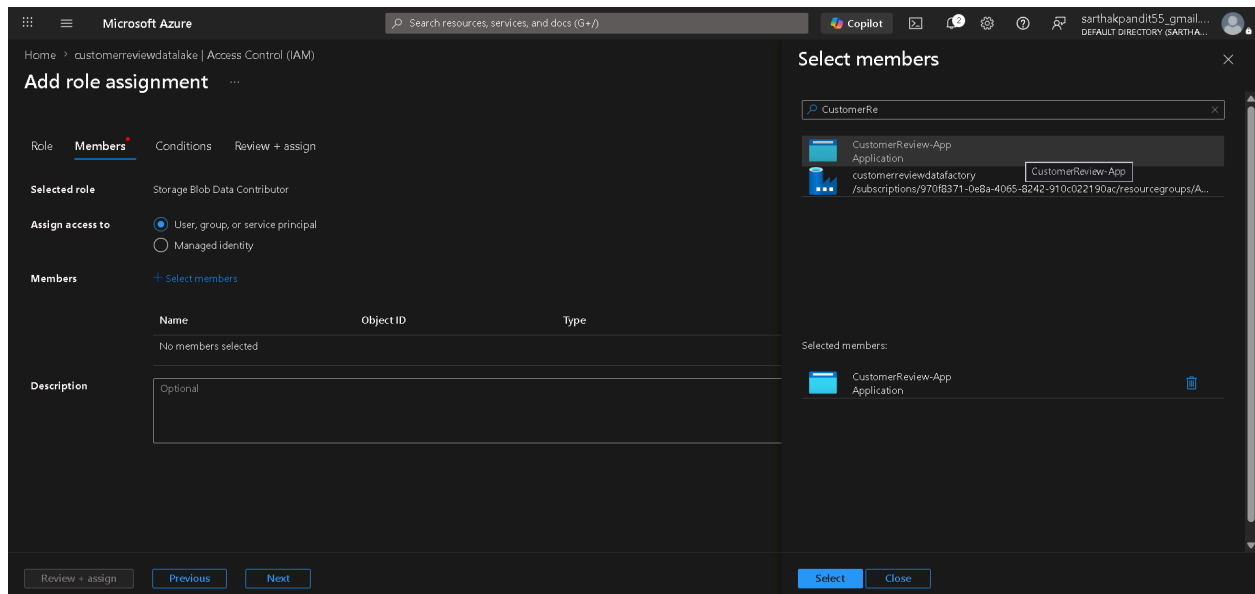
Step 8 :- Go to Storage Account “**customerreviewdatalake**” and then go to “**Access Control (IAM)**”.



Step 9 :- Click “**Add Role Assignment**”, Search “**Storage blob data contributor**” and click next.



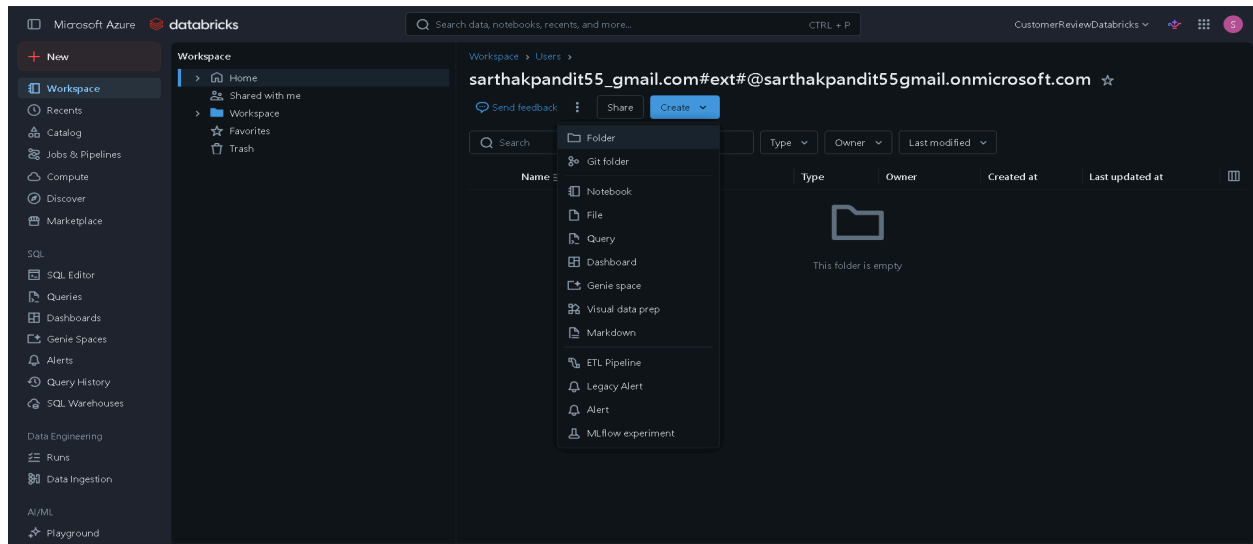
Step 10 :- Click “**Select Members**”, Select “**CustomerReview-App**” and click review + assign.



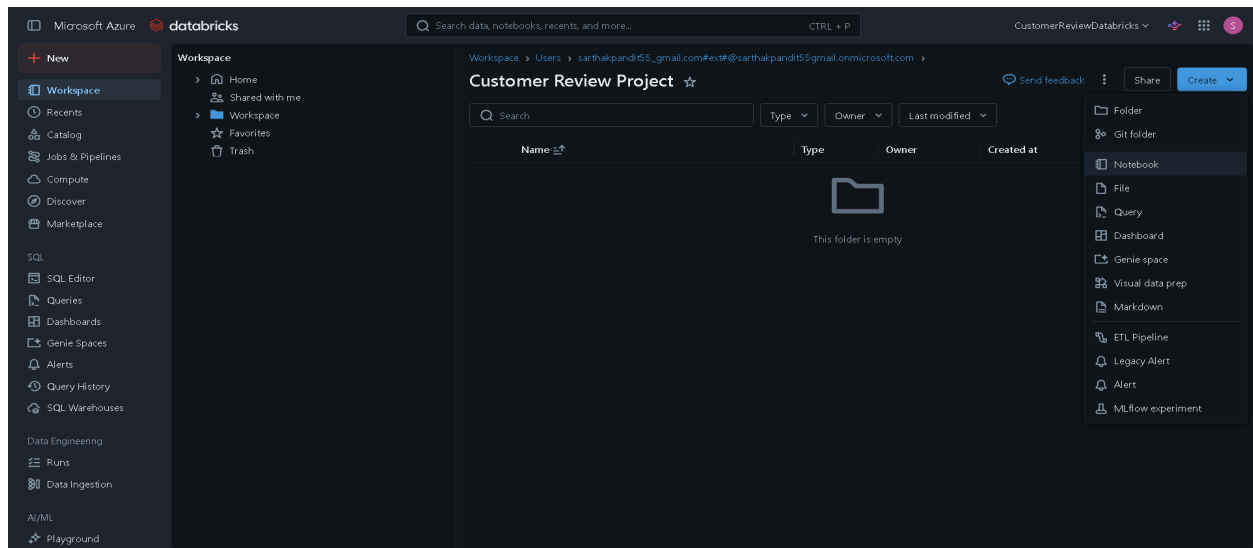
** After that the role has been assigned and can check it by going to “**Role Assignments**”.

Task4 :- Creating Databricks Notebook and Accessing Data.

Step 1 :- Go to Workspace, Click “**Create**” and create a new folder named “**Customer Review Project**”.

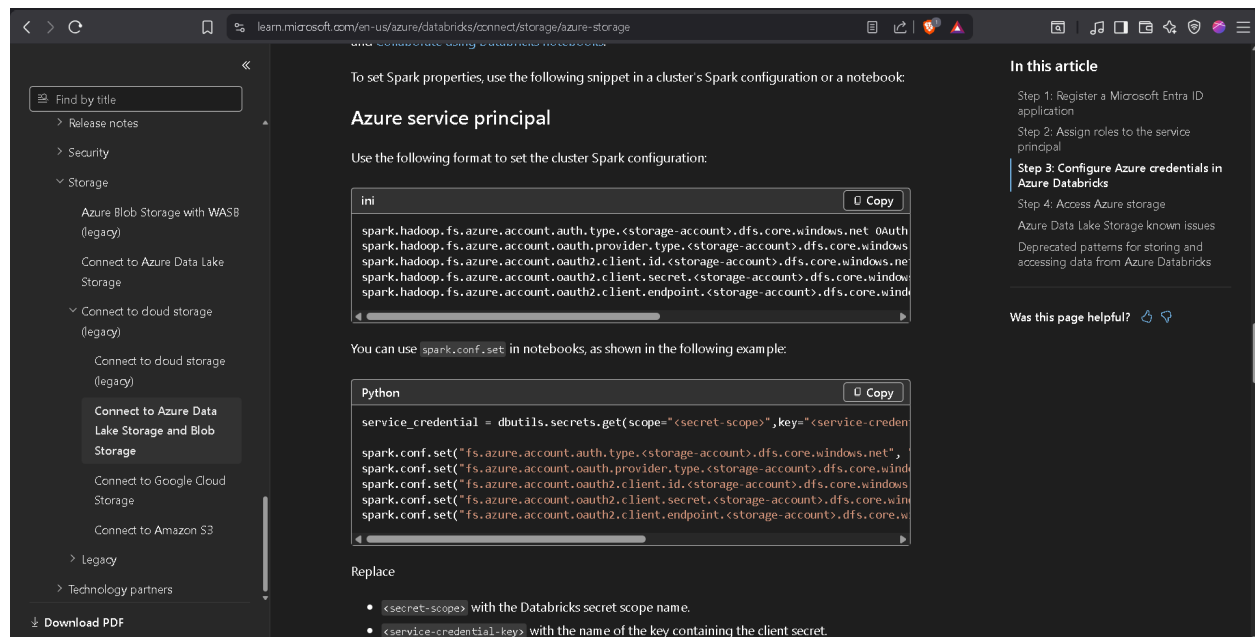


Step 2 :- Click “**Create**” and select “**Notebook**” and rename it to “**Silver_layer**”.



Step 3 :- Go to

<https://learn.microsoft.com/en-us/azure/databricks/connect/storage/azure-storage> and move to the “**Azure service principal**” and copy the commands.



The screenshot shows a web browser displaying the Microsoft Learn article "Azure service principal". The article provides instructions on how to set Spark properties for Azure storage access. It includes two code snippets: one in INI format for Spark configuration and one in Python for Databricks notebooks. The INI snippet sets various Spark properties for Azure storage access, including authentication type, OAuth provider, client ID, client secret, and endpoint. The Python snippet uses the `dbutils.secrets` API to retrieve the client secret and sets the same Spark properties. The article also includes a "Replace" section with instructions on how to substitute the placeholder values with actual Databricks secret names and keys.

To set Spark properties, use the following snippet in a cluster's Spark configuration or a notebook:

Azure service principal

Use the following format to set the cluster Spark configuration:

```
ini
spark.hadoop.fs.azure.account.auth.type.<storage-account>.dfs.core.windows.net OAuth
spark.hadoop.fs.azure.account.oauth.provider.type.<storage-account>.dfs.core.windows.net
spark.hadoop.fs.azure.account.oauth2.client.id.<storage-account>.dfs.core.windows.net
spark.hadoop.fs.azure.account.oauth2.client.secret.<storage-account>.dfs.core.windows.net
spark.hadoop.fs.azure.account.oauth2.endpoint.<storage-account>.dfs.core.windows.net
```

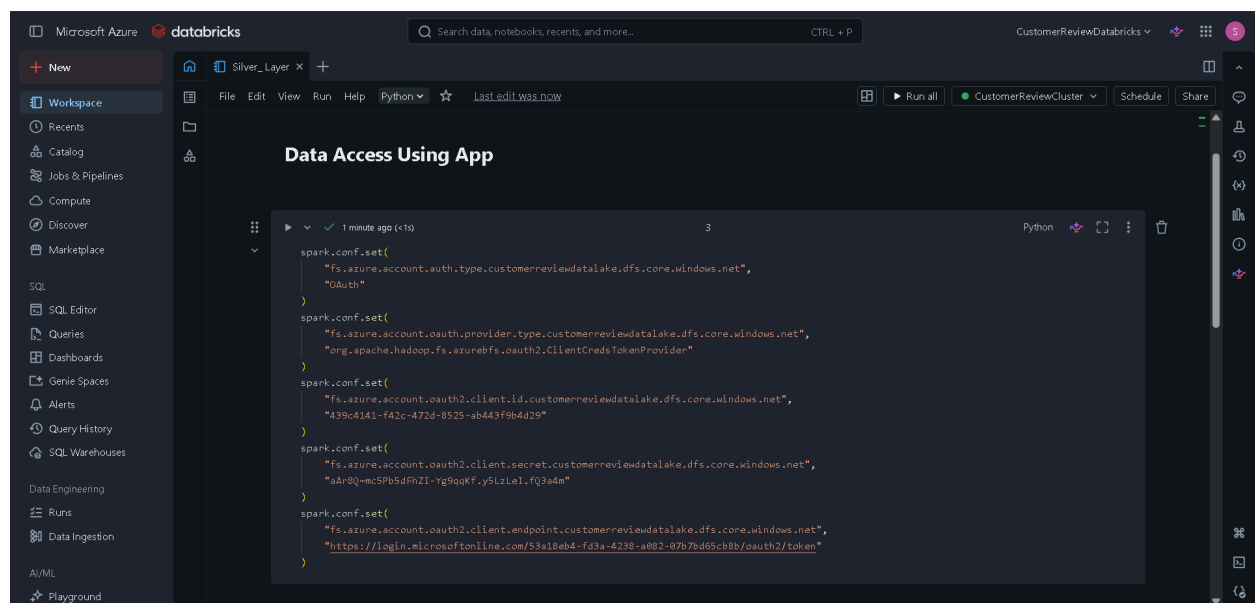
You can use `spark.conf.set` in notebooks, as shown in the following example:

```
Python
service_credential = dbutils.secrets.get(scope="<secret-scope>",key="<service-credential-key>")
spark.conf.set("fs.azure.account.auth.type.<storage-account>.dfs.core.windows.net",
spark.conf.set("fs.azure.account.oauth.provider.type.<storage-account>.dfs.core.windows.net",
spark.conf.set("fs.azure.account.oauth2.client.id.<storage-account>.dfs.core.windows.net",
spark.conf.set("fs.azure.account.oauth2.client.secret.<storage-account>.dfs.core.windows.net",
spark.conf.set("fs.azure.account.oauth2.endpoint.<storage-account>.dfs.core.windows.net",
```

Replace

- `<secret-scope>` with the Databricks secret scope name.
- `<service-credential-key>` with the name of the key containing the client secret.

Step 4 :- Paste the code and give the other application id and directory id to access the storage.



The screenshot shows the Databricks workspace interface. The left sidebar contains navigation options like "New", "Workspace", "Recents", "Catalog", "Jobs & Pipelines", "Compute", "Discover", "Marketplace", "SQL", "SQL Editor", "Queries", "Dashboards", "Genie Spaces", "Alerts", "Query History", "SQL Warehouses", "Data Engineering", "Runs", "Data Ingestion", "AI/ML", and "Playground". The main area displays a notebook titled "Data Access Using App". The notebook contains a code cell with the following Spark configuration code:

```
spark.conf.set(
  "fs.azure.account.auth.type.customerreviewdatalake.dfs.core.windows.net",
  "OAuth"
)
spark.conf.set(
  "fs.azure.account.oauth.provider.type.customerreviewdatalake.dfs.core.windows.net",
  "org.apache.hadoop.fs.azurebfs.oauth2.ClientCredsTokenProvider"
)
spark.conf.set(
  "fs.azure.account.oauth2.client.id.customerreviewdatalake.dfs.core.windows.net",
  "439c4141-f42c-472d-8525-ab443f9b4d29"
)
spark.conf.set(
  "fs.azure.account.oauth2.client.secret.customerreviewdatalake.dfs.core.windows.net",
  "sAr0Q-mc9B5dPhZI-Yg9qKf.y5LzLe1.fQ3d4m"
)
spark.conf.set(
  "fs.azure.account.oauth2.client.endpoint.customerreviewdatalake.dfs.core.windows.net",
  "https://login.microsoftonline.com/53a18ab4-fd3a-4238-a082-07b7b65cb8b8/oauth2/token"
```

Step 5 :- Testing the connection with the storage account.

The screenshot shows the Databricks interface with a notebook titled "Silver_Layer". The notebook contains a Python code snippet that tests the connection to a storage account using `dbutils.fs.is`. The code is as follows:

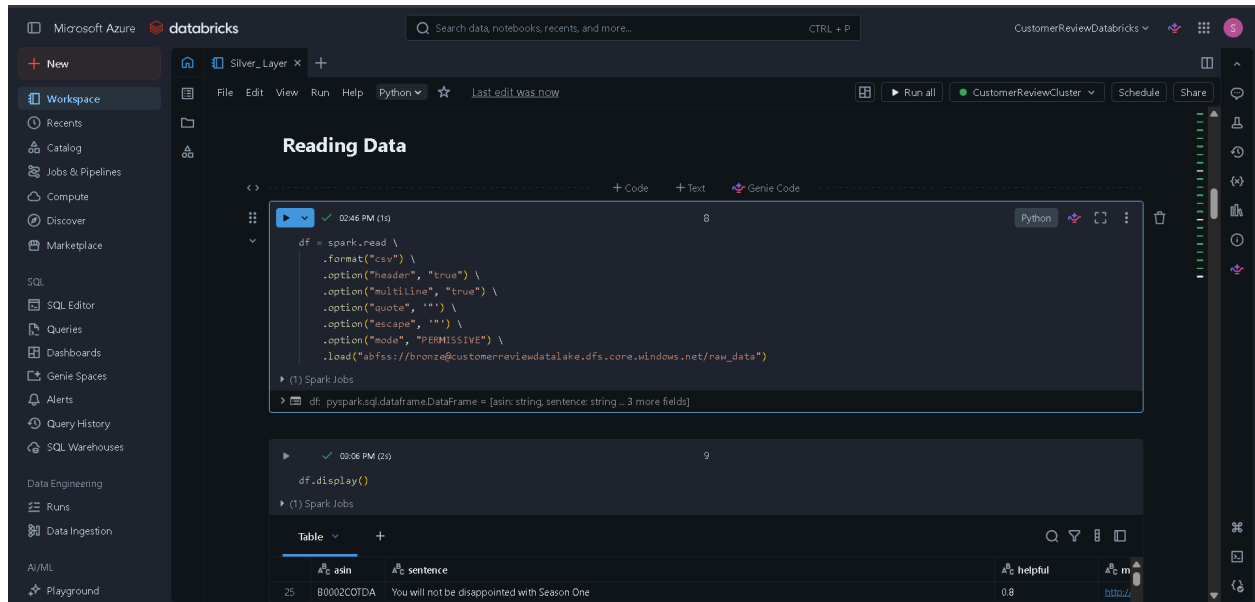
```
display(
  dbutils.fs.is(
    "abfss://bronze@customerreviewdatalake.dfs.core.windows.net/"
  )
)
```

The code has been executed successfully, as indicated by the green checkmark and the status "02:43 PM (12s)". Below the code, the output is displayed as a table with the following columns: `path`, `name`, `size`, and `modificationTime`. The table contains one row of data:

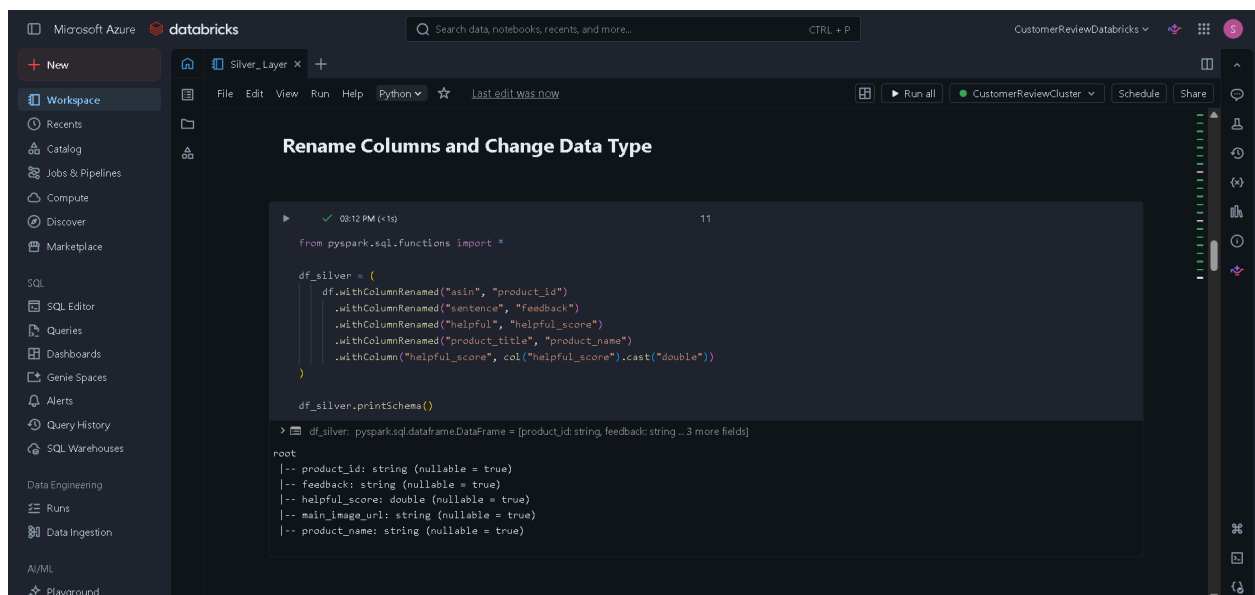
	path	name	size	modificationTime
1	abfss://bronze@customerreviewdatalake.dfs.core.windows.net/raw_data/	raw_data/	0	1782300340000

Task5 :- Performing Transformation.

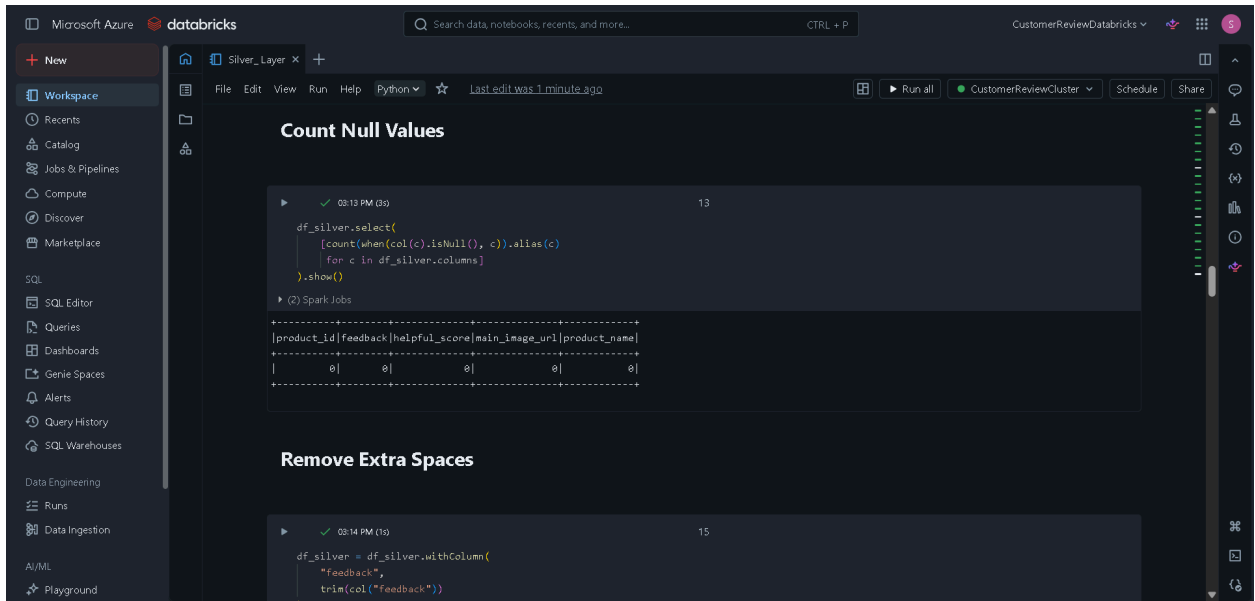
Step 1 :- Load the data from the storage account.



Step 2 :- Rename the columns and change the datatype of the column .



Step 3 :- Count the null values present in each column.



Count Null Values

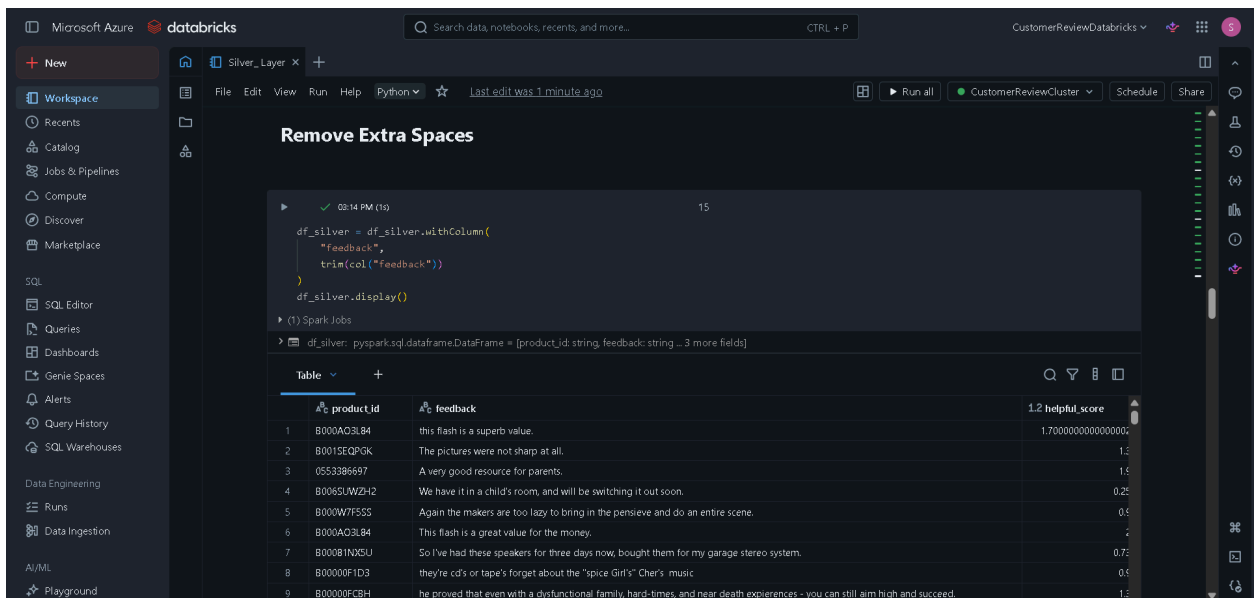
```
df_silver.select(
  [count(when(col(c).isNull(), c)).alias(c)
   for c in df_silver.columns]
).show()
```

product_id	feedback	helpful_score	main_image_url	product_name

Remove Extra Spaces

```
df_silver = df_silver.withColumn(
  "feedback",
  trim(col("feedback"))
)
```

Step 4 :- Removing Extra spaces from the “**feedback**” column.



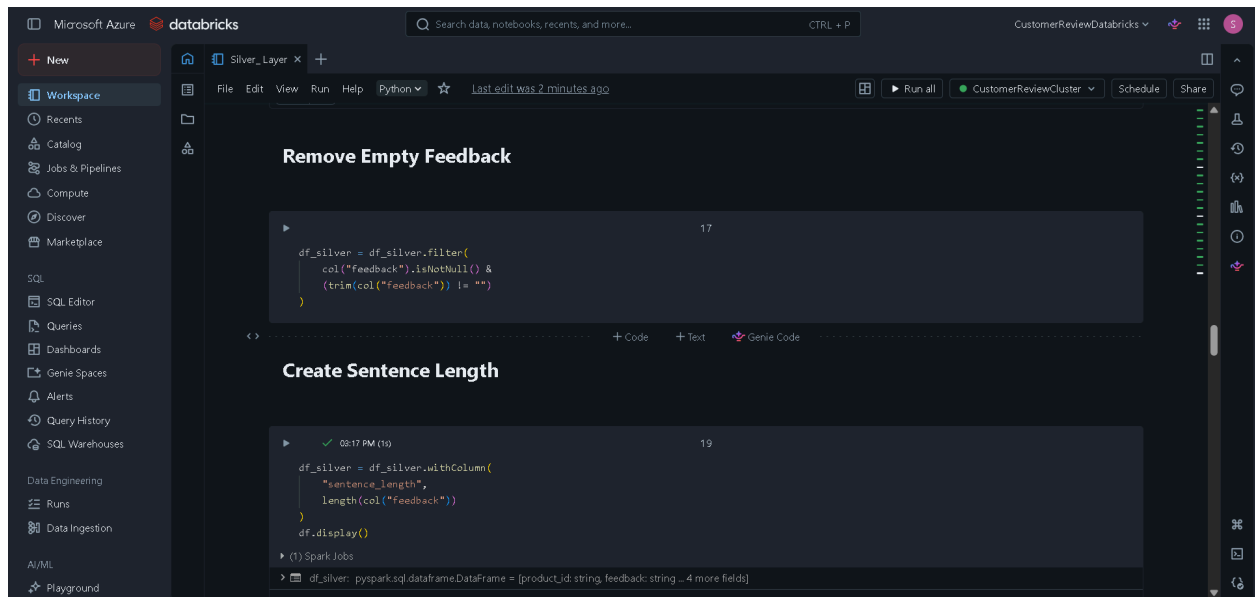
Remove Extra Spaces

```
df_silver = df_silver.withColumn(
  "feedback",
  trim(col("feedback"))
)
df_silver.display()
```

df_silver: pyspark.sql.dataframe.DataFrame = [product_id: string, feedback: string ... 3 more fields]

product_id	feedback	helpful_score
B000A03L84	this flash is a superb value.	1.7000000000000000
B0015EOPGK	The pictures were not sharp at all.	1.5
0533986697	A very good resource for parents.	1.5
B006SLWZH2	We have it in a child's room, and will be switching it out soon.	0.25
B000W7F535	Again the makers are too lazy to bring in the pensieve and do an entire scene.	0.5
B000A03L84	This flash is a great value for the money.	1.5
B00081N0SU	So I've had these speakers for three days now, bought them for my garage stereo system.	0.75
B00000F1D3	they're cd's or tape's forget about the "spice Girl's" Cher's music	0.5
B00000FCBH	he proved that even with a dysfunctional family, hard-times, and near death experiences - you can still aim high and succeed.	1.5

Step 5 :- Removing empty feedback and creating a column named “**sentence_length**” with the length of feedback.

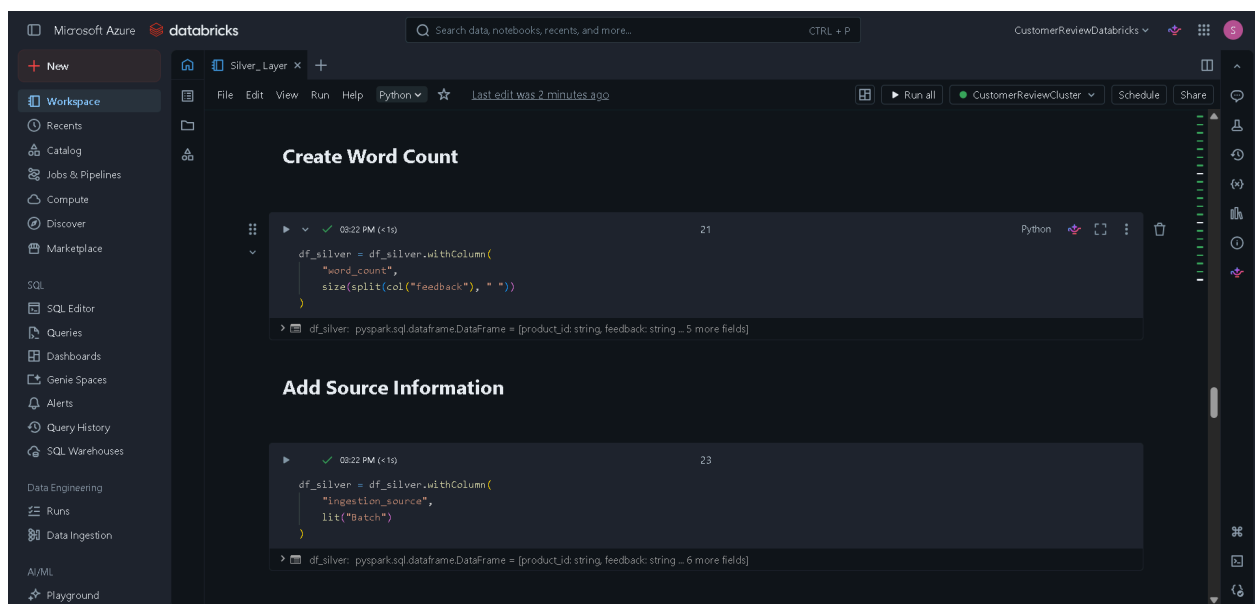


The screenshot shows the Databricks workspace interface. The left sidebar contains navigation options like Workspace, Recents, Catalog, Jobs & Pipelines, Compute, Discover, Marketplace, SQL, SQL Editor, Queries, Dashboards, Genie Spaces, Alerts, Query History, SQL Warehouses, Data Engineering, Runs, Data Ingestion, AI/ML, and Playground. The main area is titled "Silver_Layer" and shows two code blocks. The first block, titled "Remove Empty Feedback", contains Python code to filter out rows where feedback is null or empty. The second block, titled "Create Sentence Length", contains Python code to add a new column "sentence_length" based on the length of the feedback text. Both blocks show successful execution results.

```
df_silver = df_silver.filter(
    col("feedback").isNull() &
    (trim(col("feedback")) != "")
)
```

```
df_silver = df_silver.withColumn(
    "sentence_length",
    length(col("feedback"))
)
df_display()
```

Step 6 :- Create a column named “**word_count**” by finding the number of words in feedback column.

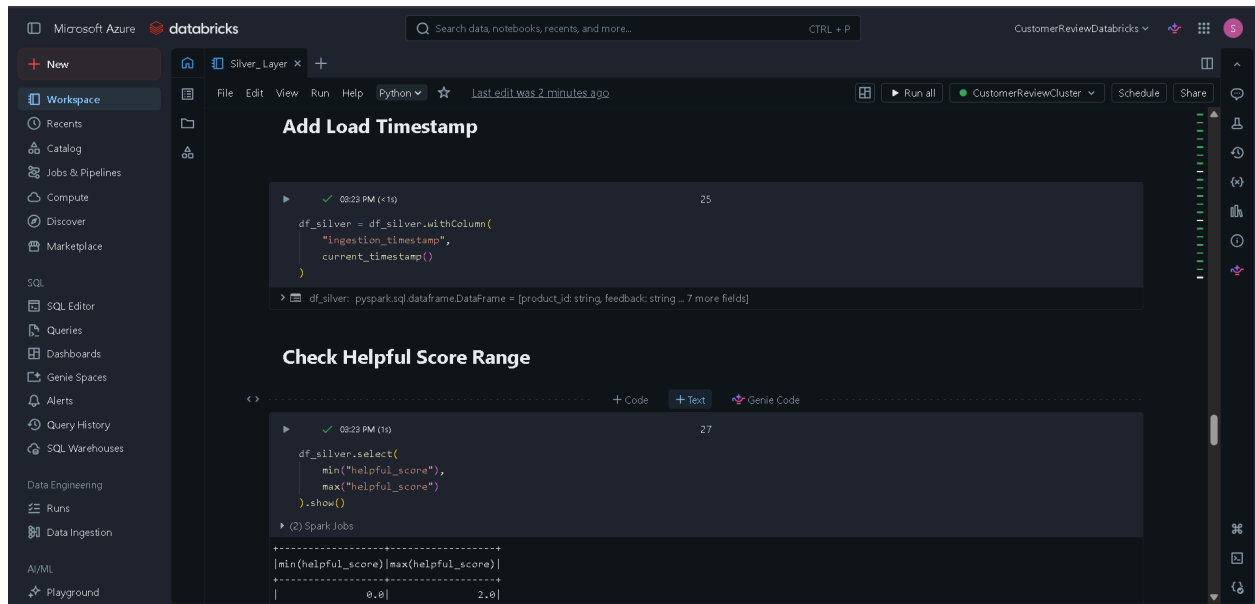


The screenshot shows the Databricks workspace interface. The left sidebar is the same as in the previous screenshot. The main area is titled "Silver_Layer" and shows two code blocks. The first block, titled "Create Word Count", contains Python code to add a new column "word_count" based on the number of words in the feedback text. The second block, titled "Add Source Information", contains Python code to add a new column "ingestion_source" with the value "Batch". Both blocks show successful execution results.

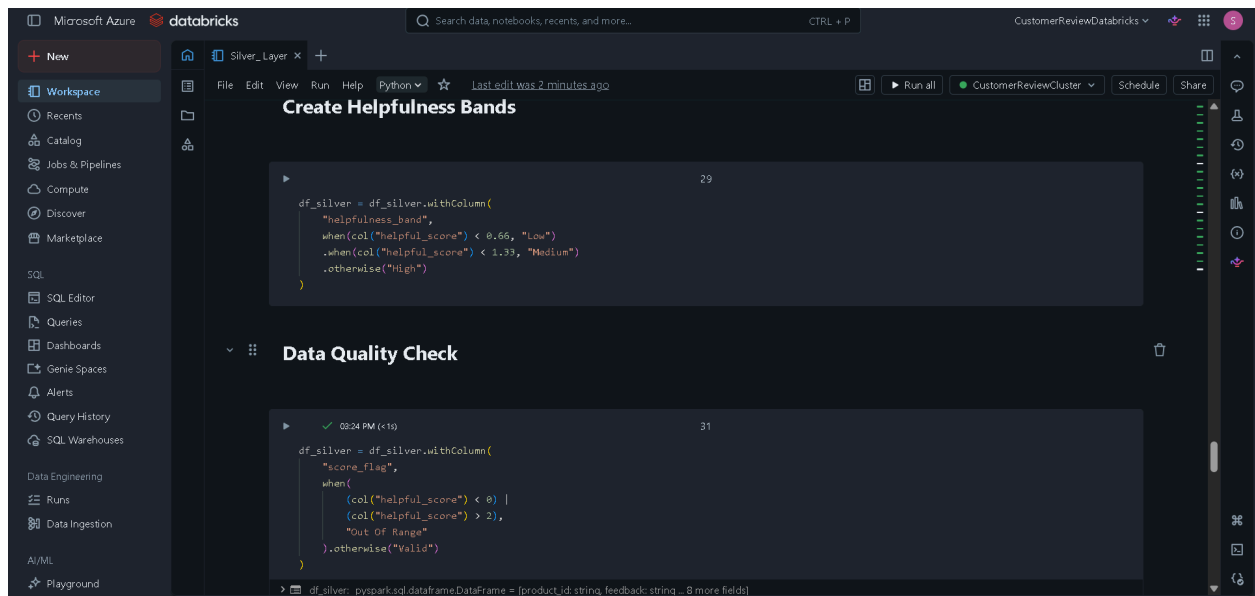
```
df_silver = df_silver.withColumn(
    "word_count",
    size(split(col("feedback"), " "))
)
```

```
df_silver = df_silver.withColumn(
    "ingestion_source",
    lit("Batch")
)
```

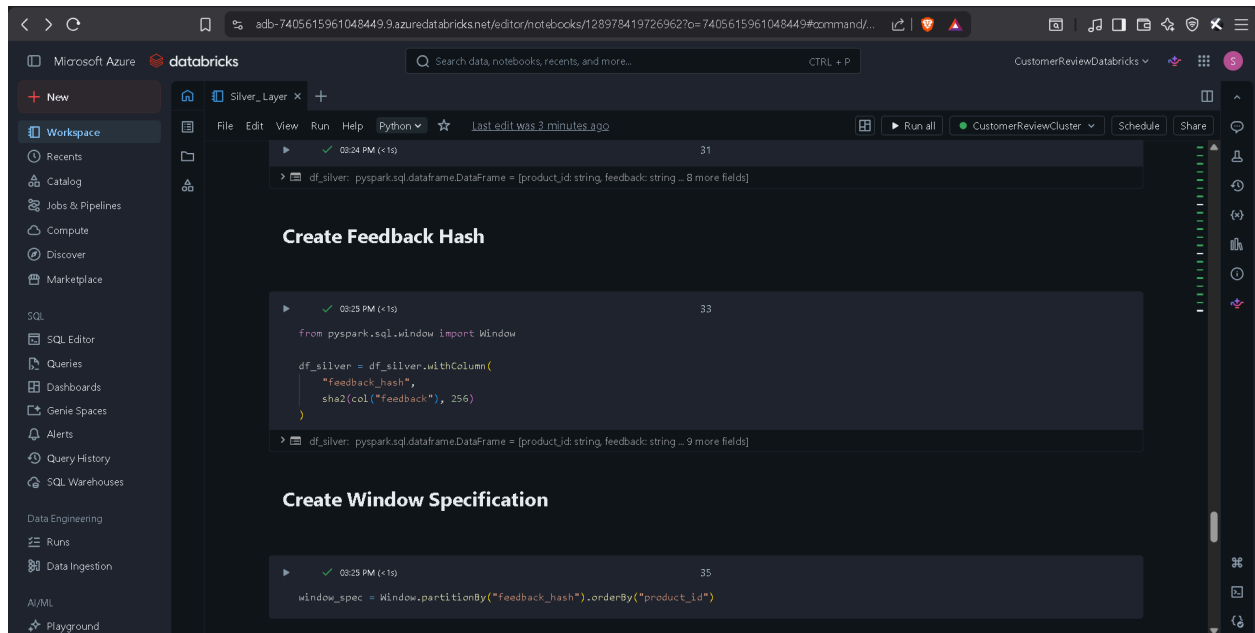
Step 7 :- Add the timestamp and find the min/max of “helpful_score”.



Step 8 :- Created the column named “helpfulness_band” applying the certain criteria.



Step 9 :- Created the “feedback_hash” column.



The screenshot shows a Databricks notebook titled "Silver_Layer" with a Python editor. The notebook is divided into three sections: "Create Feedback Hash", "Create Window Specification", and "Remove Duplicate Reviews". The "Create Feedback Hash" section contains a code block that defines a window specification and a new column "feedback_hash" using SHA2. The "Create Window Specification" section contains a code block that defines a window specification. The "Remove Duplicate Reviews" section contains a code block that filters out duplicate reviews based on the window specification. The notebook is running on a cluster named "CustomerReviewCluster".

```
df_silver: pyspark.sql.dataframe.DataFrame = [product_id: string, feedback: string ... 8 more fields]
```

Create Feedback Hash

```
from pyspark.sql.window import Window

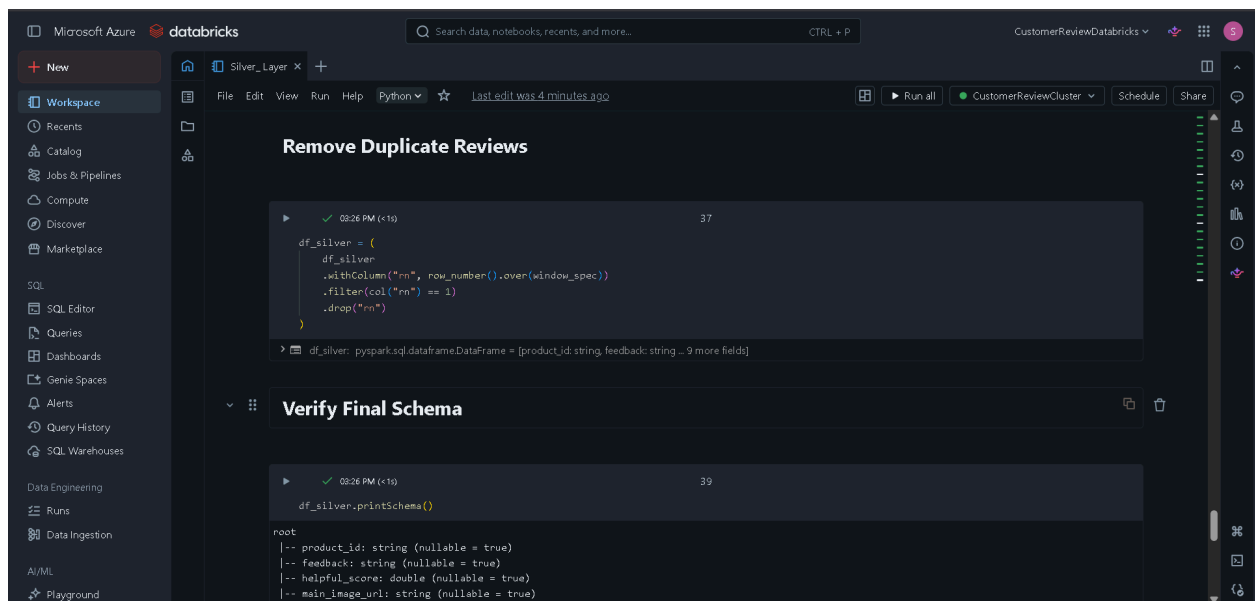
df_silver = df_silver.withColumn(
    "feedback_hash",
    sha2(col("feedback"), 256)
)
```

```
df_silver: pyspark.sql.dataframe.DataFrame = [product_id: string, feedback: string ... 9 more fields]
```

Create Window Specification

```
window_spec = Window.partitionBy("feedback_hash").orderBy("product_id")
```

Step 10 :- Removing the duplicate reviews and checking the final schema.



The screenshot shows a Databricks notebook titled "Silver_Layer" with a Python editor. The notebook is divided into three sections: "Remove Duplicate Reviews", "Verify Final Schema", and "Remove Duplicate Reviews". The "Remove Duplicate Reviews" section contains a code block that filters out duplicate reviews based on the window specification. The "Verify Final Schema" section contains a code block that prints the schema of the resulting DataFrame. The notebook is running on a cluster named "CustomerReviewCluster".

```
df_silver = {
    df_silver
    .withColumn("rn", row_number().over(window_spec))
    .filter(col("rn") == 1)
    .drop("rn")
}
```

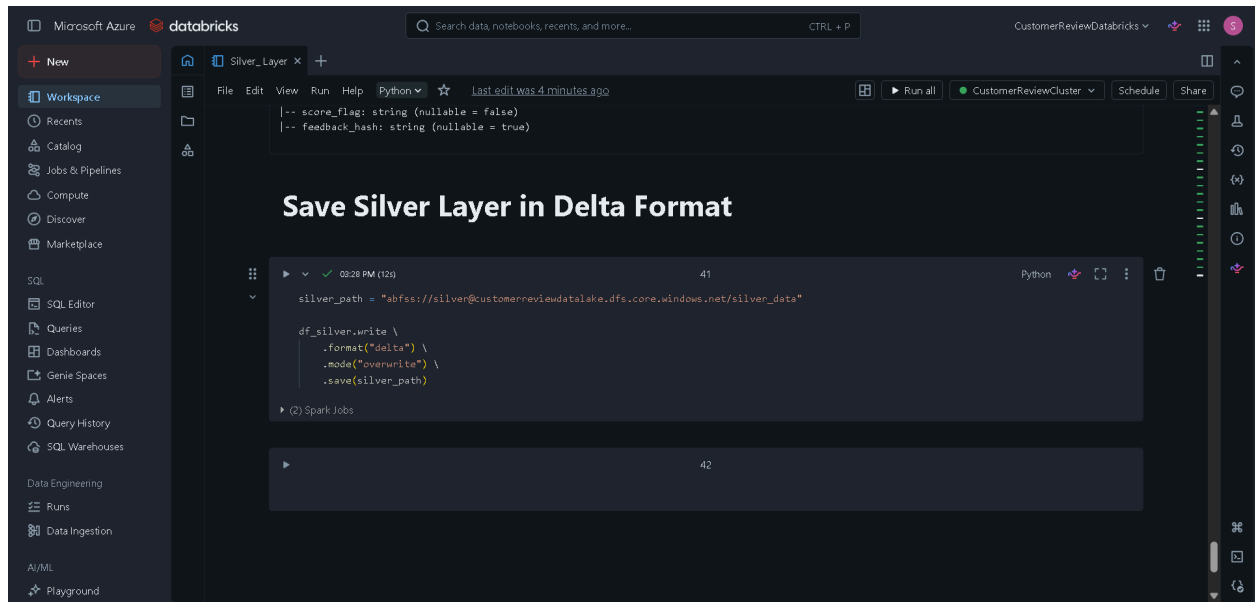
```
df_silver: pyspark.sql.dataframe.DataFrame = [product_id: string, feedback: string ... 9 more fields]
```

Verify Final Schema

```
df_silver.printSchema()
```

```
root
|-- product_id: string (nullable = true)
|-- feedback: string (nullable = true)
|-- helpful_score: double (nullable = true)
|-- main_image_url: string (nullable = true)
```

Step 11:- Save the data in “**silver_data**” after the transformation.



Phase-3 (Gold)

Task1 :- Creating Azure Synapse Analytics.

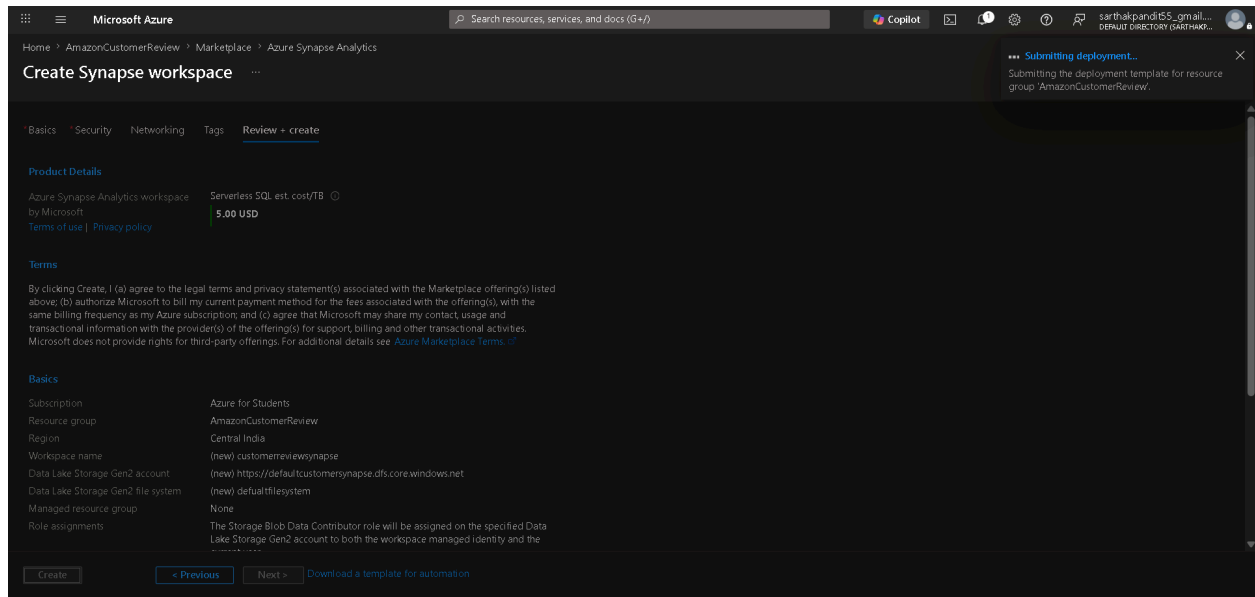
Step 1 :- Create Azure Synapse Analytics “customerreviewsynapse”.

The screenshot shows the 'Create Synapse workspace' wizard in the Azure portal, specifically the 'Basics' tab. The breadcrumb navigation at the top reads 'Home > AmazonCustomerReview > Marketplace > Azure Synapse Analytics'. The page title is 'Create Synapse workspace'. Below the title are tabs for 'Basics', 'Security', 'Networking', 'Tags', and 'Review + create'. A brief instruction states: 'Create a Synapse workspace to develop an enterprise analytics solution in just a few clicks.' The 'Project details' section asks to 'Select the subscription to manage deployed resources and costs. Use resource groups like folders to organize and manage all of your resources.' It includes three dropdowns: 'Subscription' (set to 'Azure for Students'), 'Resource group' (set to 'AmazonCustomerReview'), and 'Managed resource group' (with a placeholder 'Enter managed resource group name'). The 'Workspace details' section asks to 'Name your workspace, select a location, and choose a primary Data Lake Storage Gen2 file system to serve as the default location for logs and job output.' It includes four dropdowns: 'Workspace name' (set to 'customerreviewsynapse'), 'Region' (set to '(Asia Pacific) Central India'), 'Select Data Lake Storage Gen2' (with radio buttons for 'From subscription' and 'Manually via URL', where 'From subscription' is selected), and 'Account name' (set to '(New) defaultcustomersynapse'). Below these are two more dropdowns: 'File system name' (set to '(New) defaultfilesystem') and 'Create new' links for each. At the bottom are three buttons: 'Review + create', '< Previous', and 'Next Security >'.

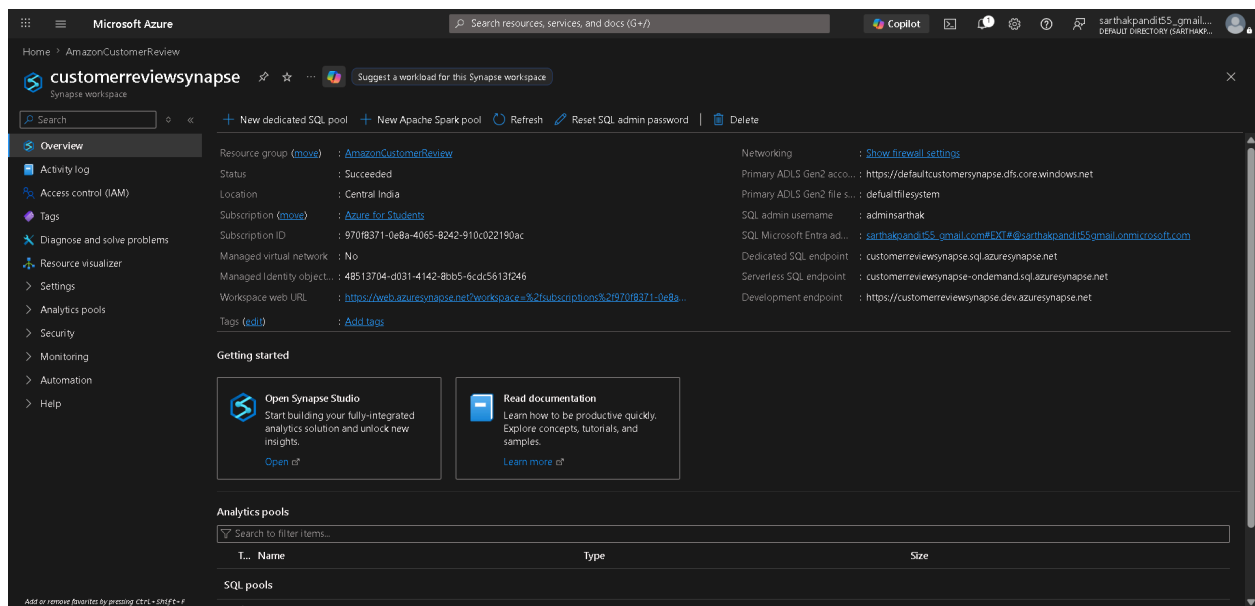
Step 2 :- Set the Sql server admin and password.

The screenshot shows the 'Create Synapse workspace' wizard in the Azure portal, specifically the 'Security' tab. The breadcrumb navigation at the top reads 'Home > AmazonCustomerReview > Marketplace > Azure Synapse Analytics'. The page title is 'Create Synapse workspace'. Below the title are tabs for 'Basics', 'Security', 'Networking', 'Tags', and 'Review + create'. A brief instruction states: 'Configure security options for your workspace.' The 'Authentication' section asks to 'Choose the authentication method for access to workspace resources such as SQL pools. The authentication method can be changed later on. [Learn more](#)'. It includes two radio buttons: 'Use both local and Microsoft Entra ID authentication' (selected) and 'Use only Microsoft Entra ID authentication'. Below these are three input fields: 'SQL Server admin login' (set to 'adminsarthak'), 'SQL Password' (masked with dots), and 'Confirm password' (masked with dots). The 'System assigned managed identity permission' section asks to 'Select to grant the workspace network access to the Data Lake Storage Gen2 account using the workspace system identity. [Learn more](#)'. It includes a checkbox 'Allow network access to Data Lake Storage Gen2 account.' which is unchecked. Below this is an information icon and a message: 'The selected Data Lake Storage Gen2 account does not restrict network access using any network access rules or you selected a storage account manually via URL under Basics tab. [Learn more](#)'. The 'Workspace encryption' section has a warning icon and a message: 'Double encryption configuration cannot be changed after opting into using a customer-managed key at the time of...'. At the bottom are three buttons: 'Review + create', '< Previous', and 'Next: Networking >'.

Step 3 :- Click “**Create**” and create the synapse analytics.

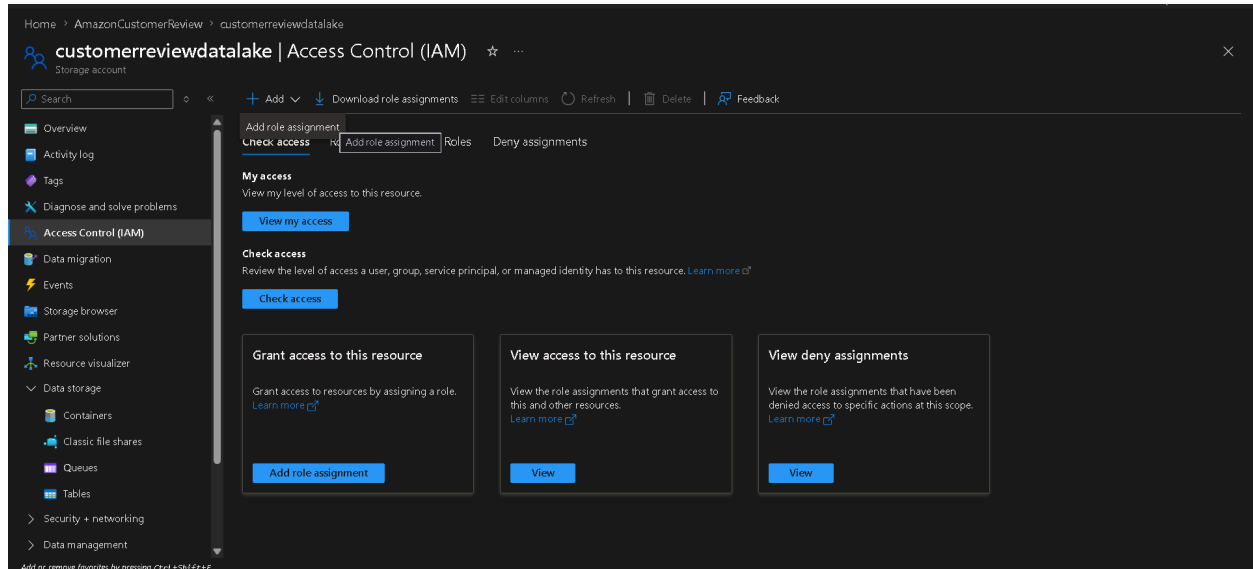


Step 4 :- Synapse Analytics is successfully created.

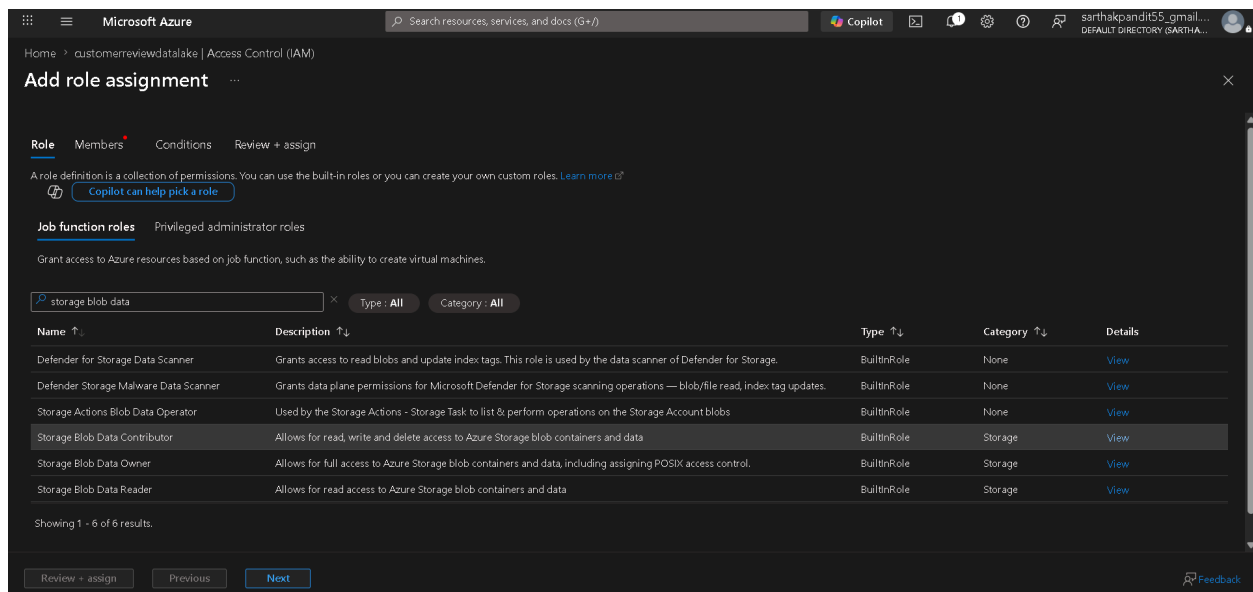


Task2 :- Providing Data Storage access to Azure Synapse Analytics.

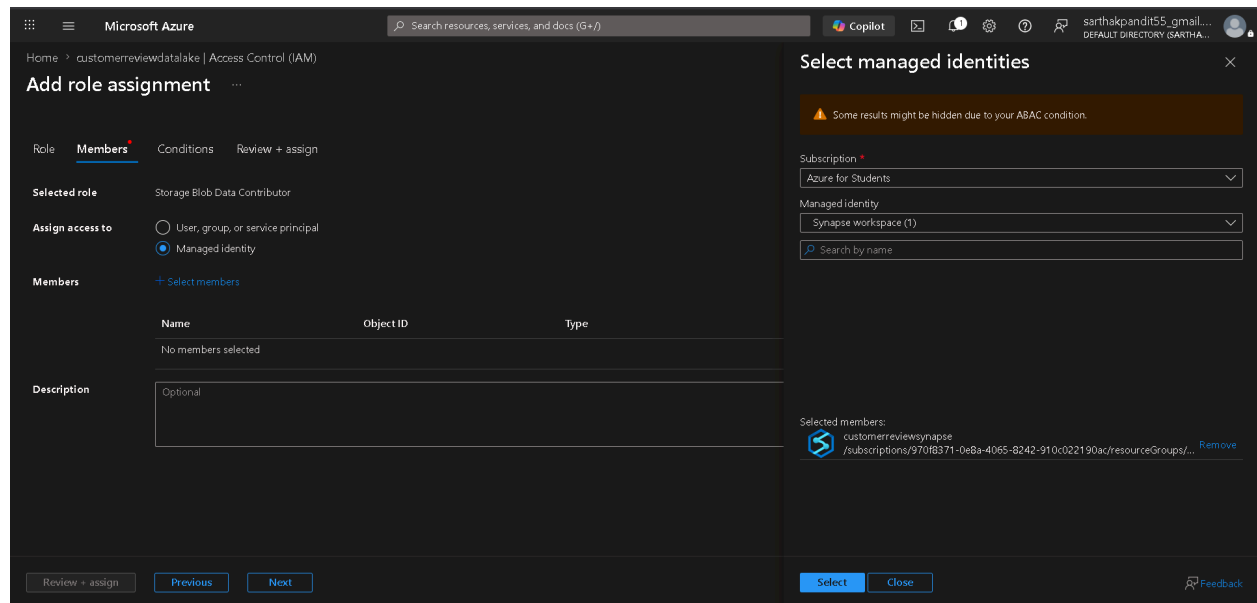
Step 1 :- Go to “customerreviewdatalake” Storage Account then to “Access Control (IAM)” and click “Add new assignment”.



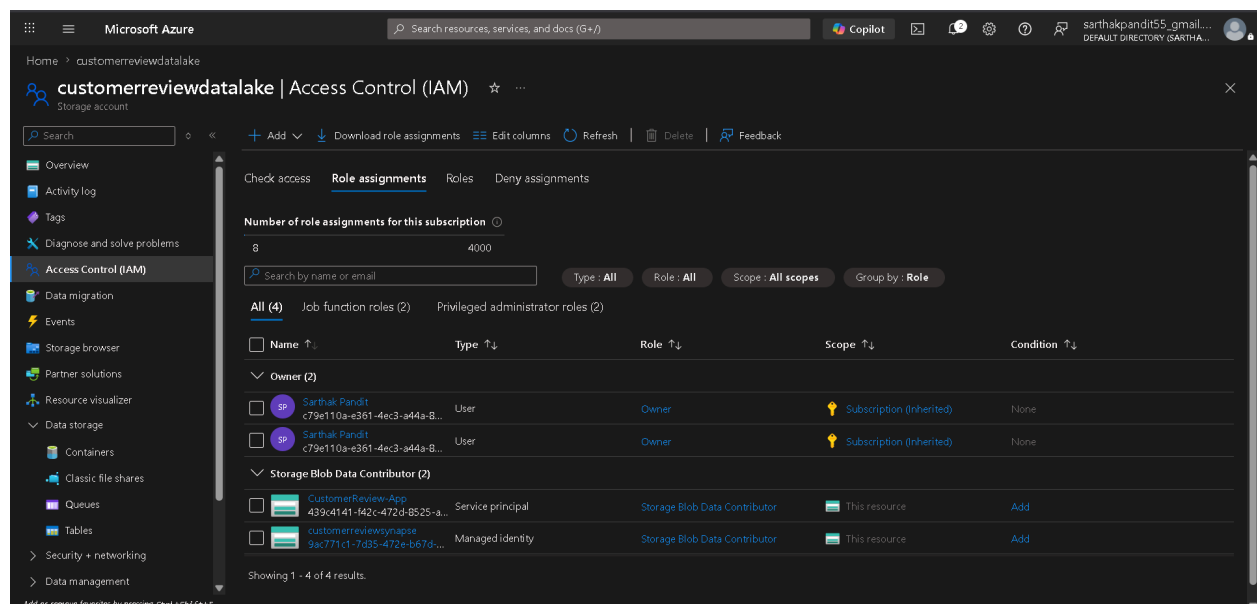
Step 2 :- Search and select “storage blob data contributor” then click next.



Step 3 :- Select “**Managed Identity**” then click “**Select Member**” and select “**customerreviewsynapse**”.

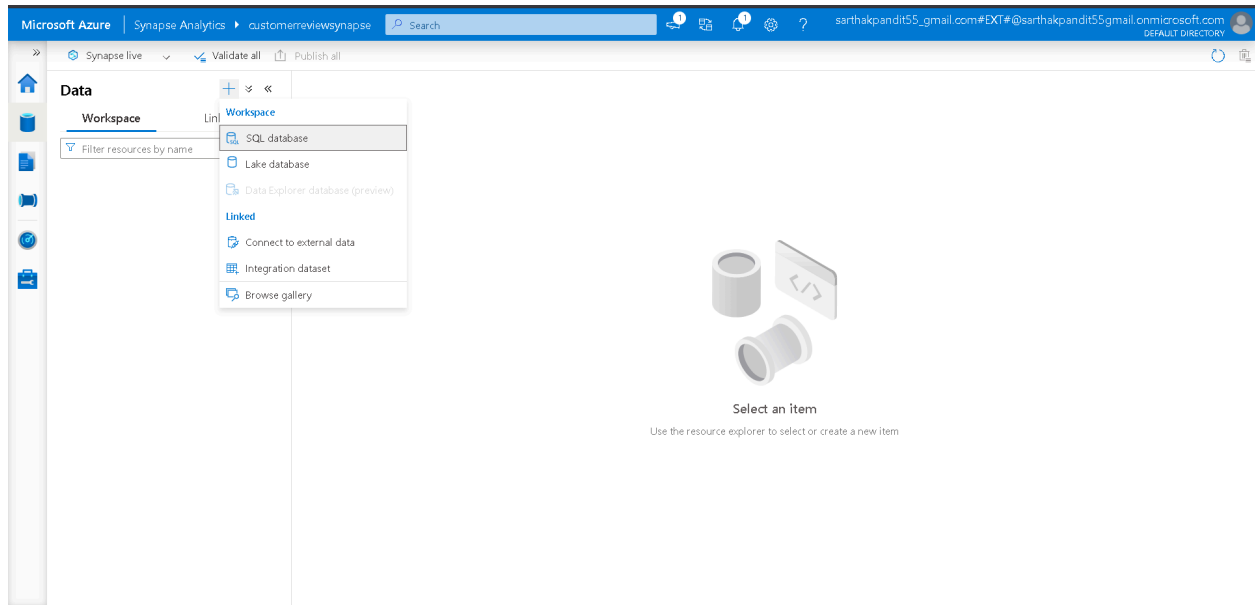


Step 4 :- Role is successfully assigned.

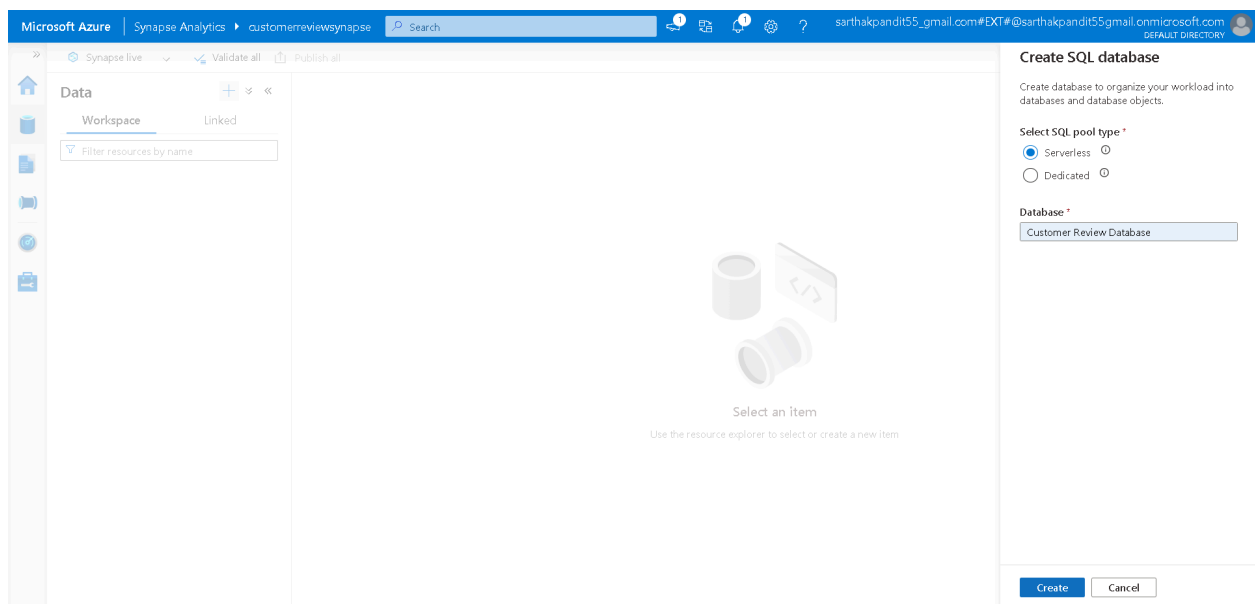


Task3 :- Create Database.

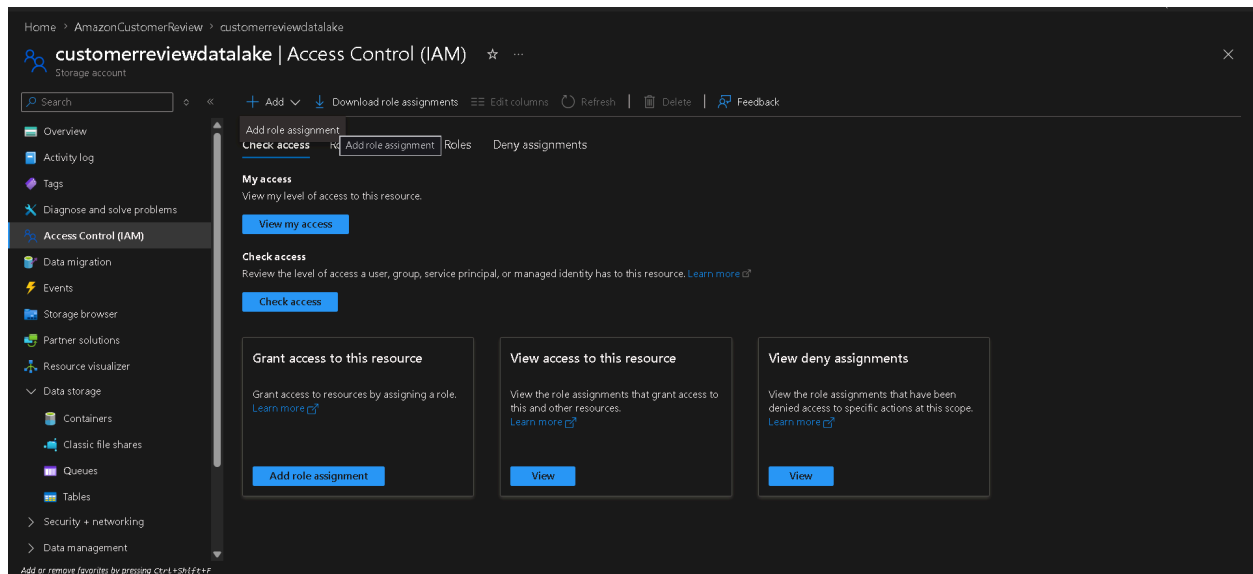
Step 1 :- Go to the “**Data**” tab and click create “**SQL Database**”.



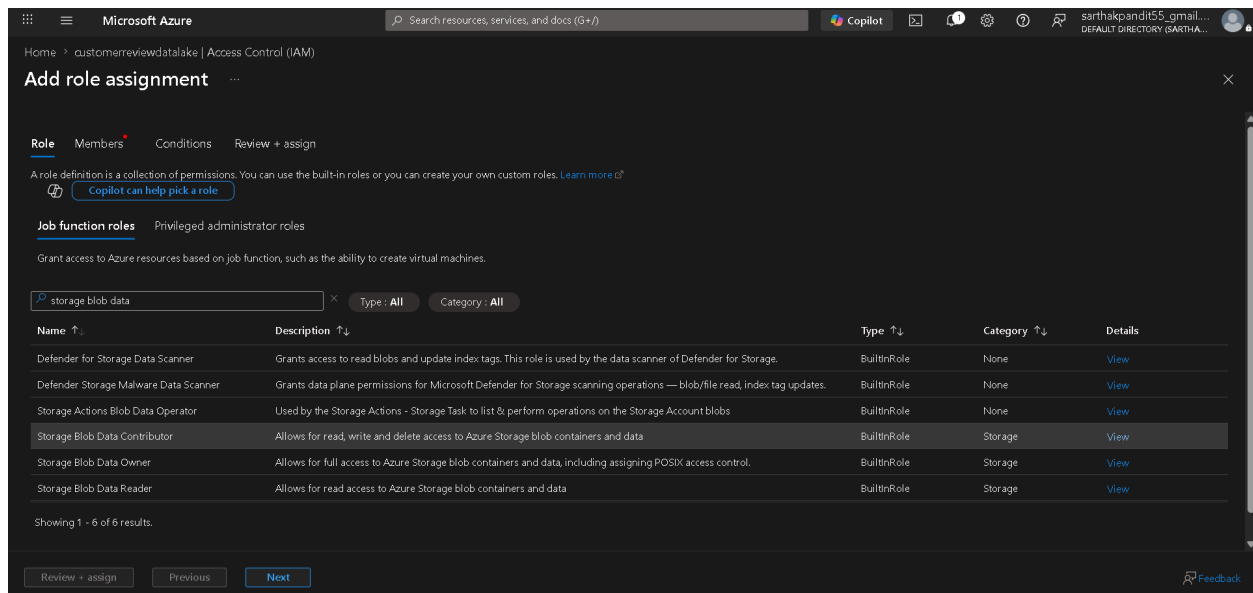
Step 2 :- Select “**Serverless**” and enter the database name and create.



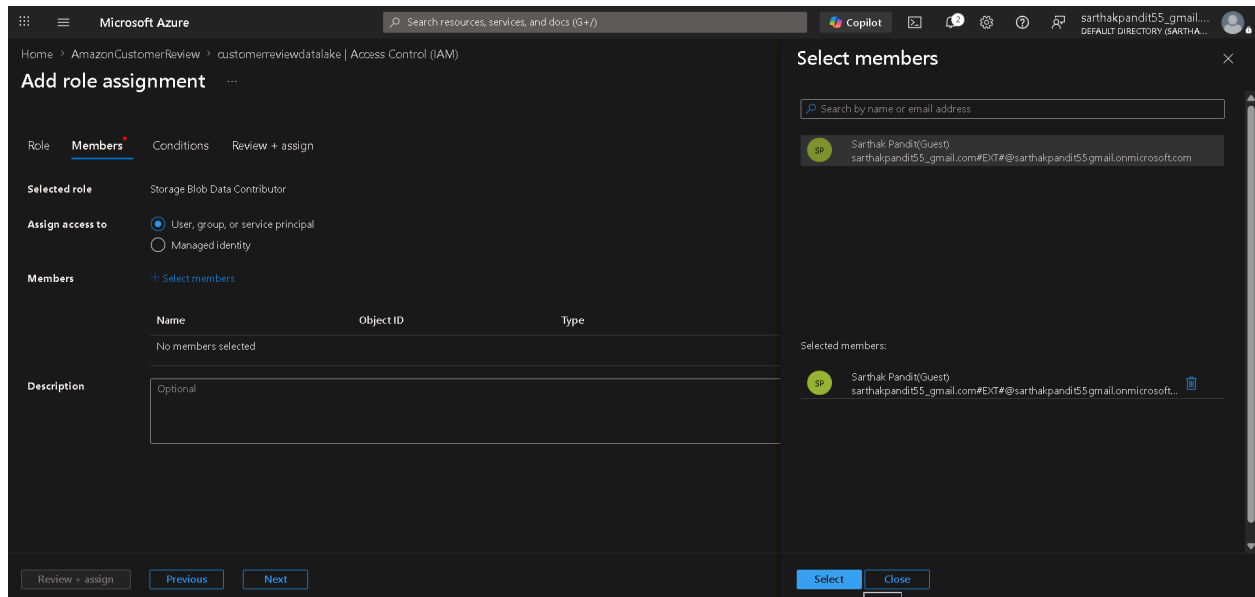
Step 3 :- Go to “customerreviewdatalake” Storage Account then to “Access Control (IAM)” and click “Add new assignment”.



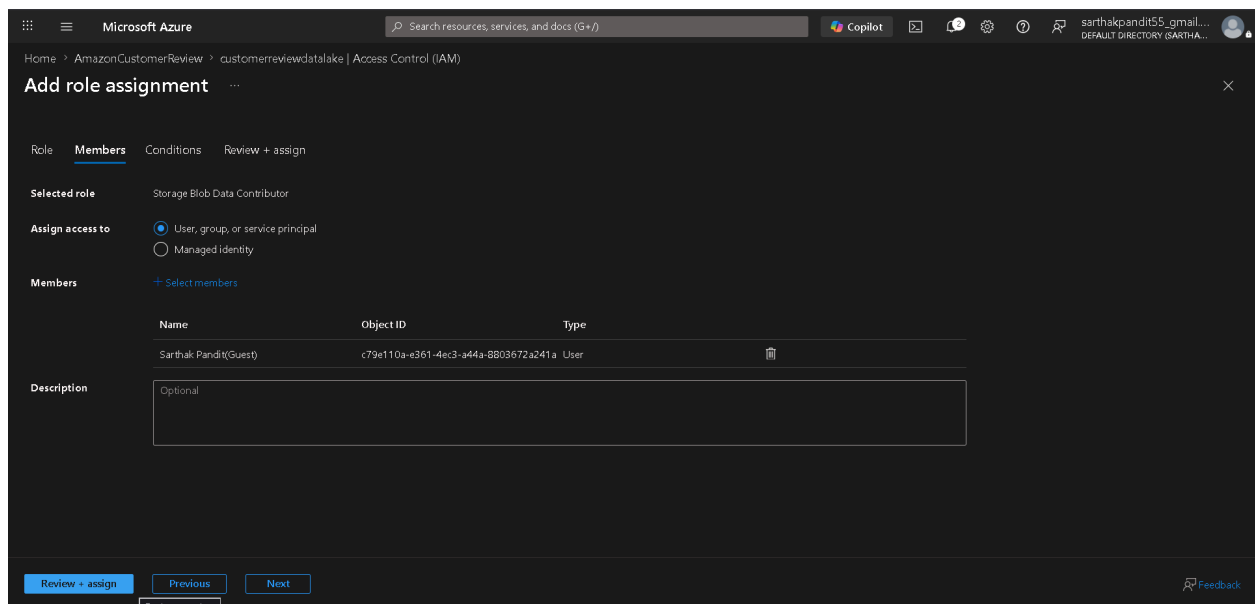
Step 4 :- Search and select “storage blob data contributor” then click next.



Step 5 :- Select the User ID.

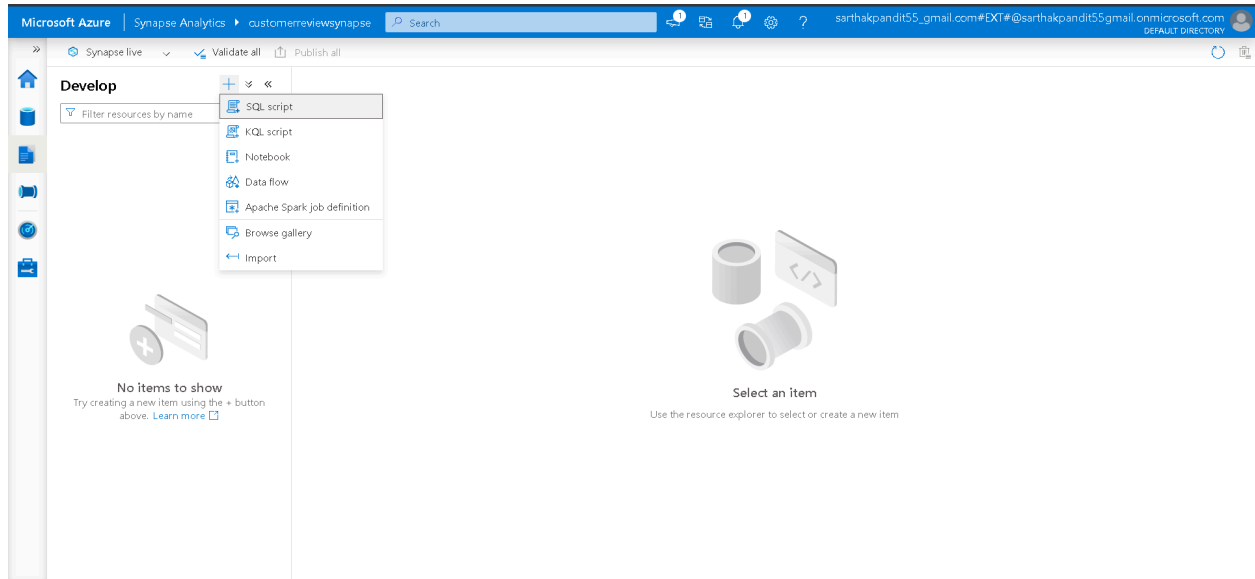


Step 6 :- Click “**Review + Create**” the role is assigned.

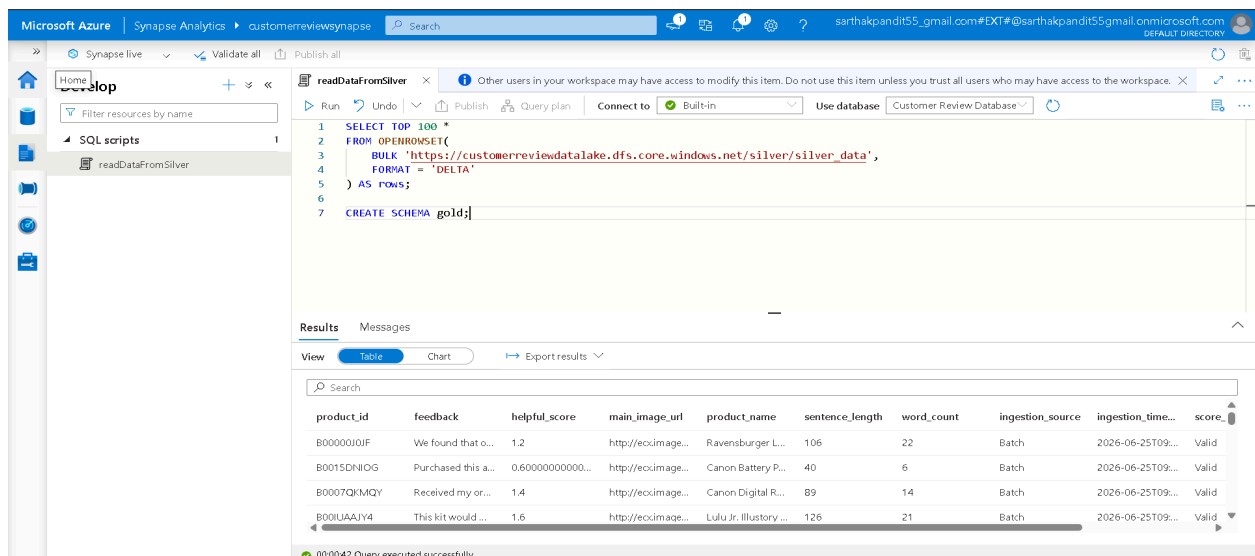


Task4 :- Writing Business KPIs / Business Use Cases.

Step 1 :- Go to the “**Develop**” tab and click create “**SQL Script**” and rename it “**readDataFromSilver**”.



Step 2 :- Write the SQL Script to read the data from silver and create Schema.



Step 3 :- Write the SQL Script to create “View” in “viewCreationGold” script.

The screenshot shows the Microsoft Azure Synapse Analytics interface. The left sidebar displays the 'Develop' environment with a list of SQL scripts: 'readDataFromSilver' and 'viewCreationGold'. The main editor window shows the 'viewCreationGold' script being executed. The script contains the following SQL code:

```

1  --Cerating View
2
3  CREATE OR ALTER VIEW gold.vw_review_silver
4  AS
5  SELECT *
6  FROM OPENROWSET(
7    BULK 'https://customerreviewdatalake.dfs.core.windows.net/silver/silver_data',
8    FORMAT = 'DELTA'
9  ) AS rows;
10
11 --Showing the View
12 SELECT TOP 10 * FROM gold.vw_review_silver;

```

The 'Results' pane shows the output of the query, displaying a table with 10 columns: product_id, feedback, helpful_score, main_image_url, product_name, sentence_length, word_count, ingestion_source, ingestion_time..., and score. The table contains two rows of data.

product_id	feedback	helpful_score	main_image_url	product_name	sentence_length	word_count	ingestion_source	ingestion_time...	score
B00000J0JF	We found that o...	1.2	http://ecximage...	Ravensburger L...	106	22	Batch	2026-06-25T09:...	Valid
B0015DNIOG	Purchased this a...	0.600000000000...	http://ecximage...	Canon Battery P...	40	6	Batch	2026-06-25T09:...	Valid

The status bar at the bottom indicates '00:00:02 Query executed successfully.'

Step 4 :- Create a folder “GoldLayerPopulate”. In this folder create a file for use case 1 named “UseCase-1” and write the query.

The screenshot shows the Microsoft Azure Synapse Analytics interface. The left sidebar displays the 'Develop' environment with a list of SQL scripts: 'readDataFromSilver', 'viewCreationGold', 'GoldLayerPopulate', and 'UseCase-1'. The main editor window shows the 'UseCase-1' script being executed. The script contains the following SQL code:

```

1  -- Use Case 1: Helpfulness Segmentation
2
3  CREATE OR ALTER VIEW gold.HelpfulnessSegments
4  AS
5  SELECT
6    helpfulness_band,
7    COUNT(*) AS review_count,
8    AVG(helpful_score) AS avg_helpful_score,
9    AVG(sentence_length) AS avg_sentence_length,
10   AVG(word_count) AS avg_word_count
11 FROM gold.vw_review_silver
12 GROUP BY helpfulness_band;
13 GO
14
15 SELECT * FROM gold.HelpfulnessSegments;

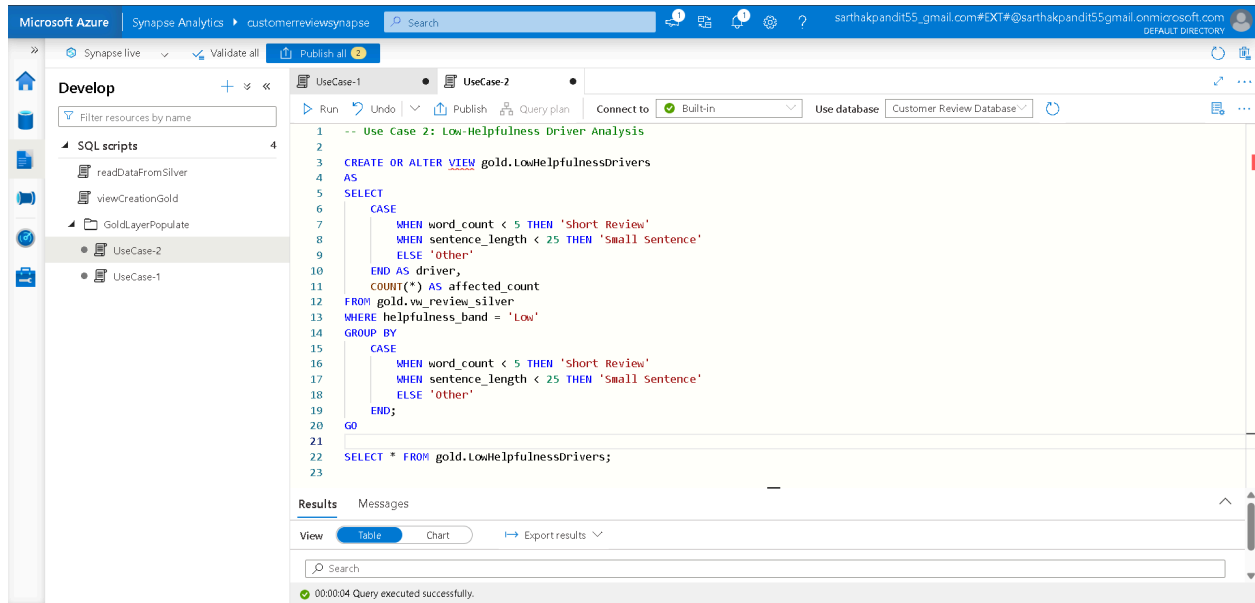
```

The 'Results' pane shows the output of the query, displaying a table with 5 columns: helpfulness_band, review_count, avg_helpful_score, avg_sentence_length, and avg_word_count. The table contains three rows of data.

helpfulness_band	review_count	avg_helpful_score	avg_sentence_length	avg_word_count
Low	2452	0.44655383360522005	50	9
Medium	10453	1.0426652635606988	73	13
High	7095	1.5818661028893628	88	16

The status bar at the bottom indicates '00:00:03 Query executed successfully.'

Step 5 :- Now Create SQL Script file for use case 2 named **“UseCase-2”** and write the script.



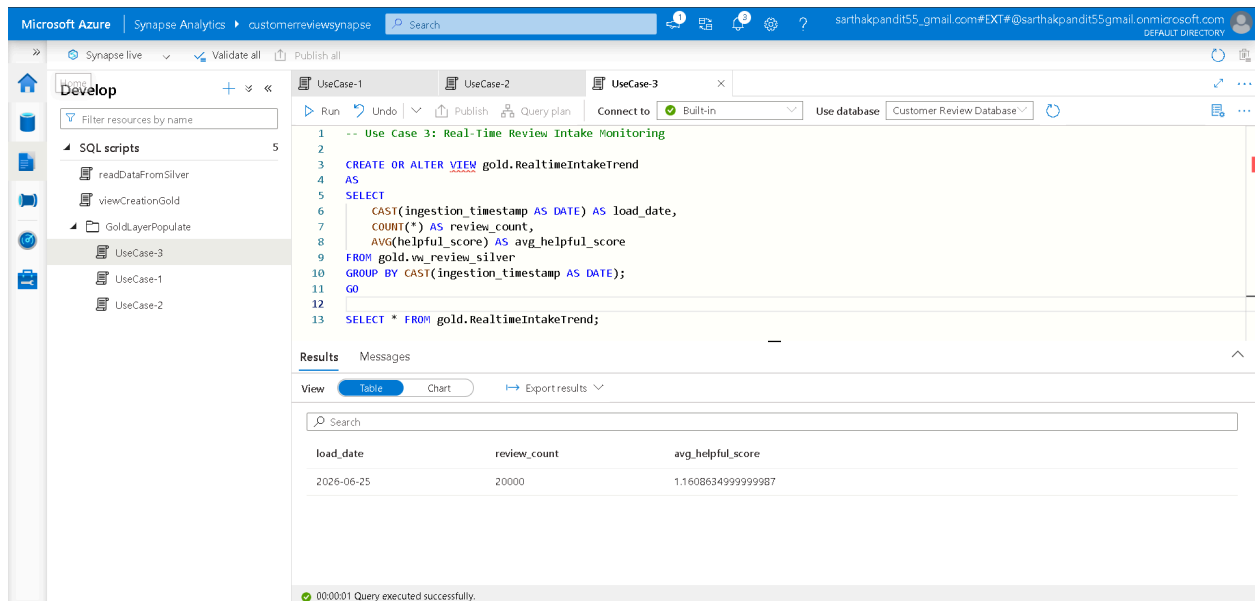
```
1 -- Use Case 2: Low-Helpfulness Driver Analysis
2
3 CREATE OR ALTER VIEW gold.LowHelpfulnessDrivers
4 AS
5 SELECT
6     CASE
7         WHEN word_count < 5 THEN 'Short Review'
8         WHEN sentence_length < 25 THEN 'Small Sentence'
9         ELSE 'other'
10    END AS driver,
11    COUNT(*) AS affected_count
12 FROM gold.wv_review_silver
13 WHERE helpfulness_band = 'Low'
14 GROUP BY
15     CASE
16         WHEN word_count < 5 THEN 'Short Review'
17         WHEN sentence_length < 25 THEN 'Small Sentence'
18         ELSE 'other'
19    END;
20 GO
21
22 SELECT * FROM gold.LowHelpfulnessDrivers;
23
```

Results Messages

View Table Chart Export results

000004 Query executed successfully.

Step 6 :- Now Create SQL Script file for use case 3 named **“UseCase-3”** and write the script.



```
1 -- Use Case 3: Real-Time Review Intake Monitoring
2
3 CREATE OR ALTER VIEW gold.RealtimeIntakeTrend
4 AS
5 SELECT
6     CAST(ingestion_timestamp AS DATE) AS load_date,
7     COUNT(*) AS review_count,
8     AVG(helpful_score) AS avg_helpful_score
9 FROM gold.wv_review_silver
10 GROUP BY CAST(ingestion_timestamp AS DATE);
11 GO
12
13 SELECT * FROM gold.RealtimeIntakeTrend;
```

Results Messages

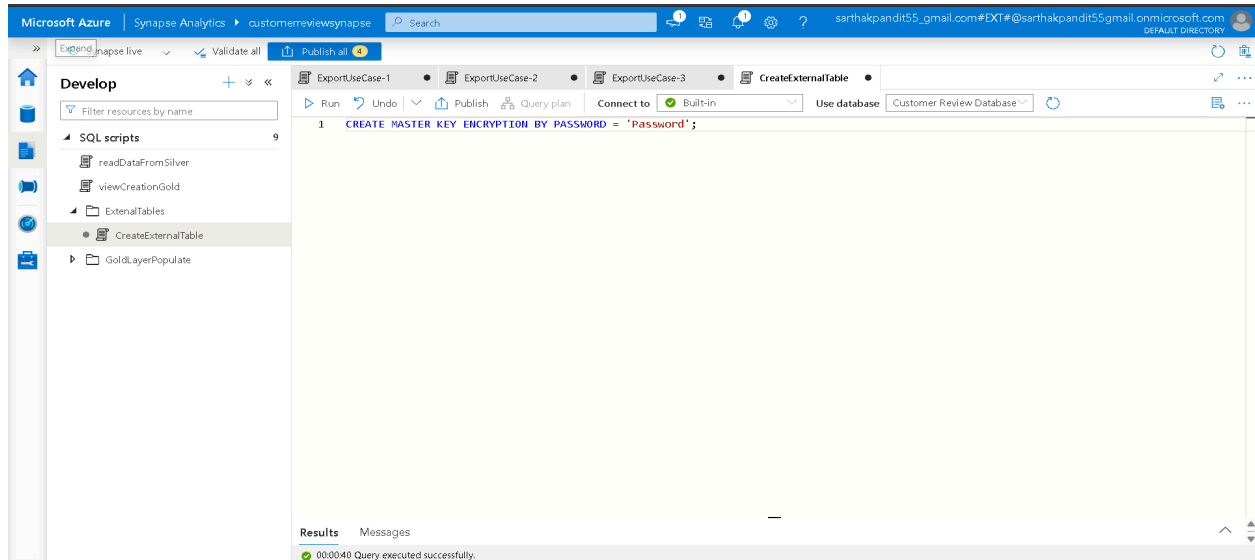
View Table Chart Export results

load_date	review_count	avg_helpful_score
2026-06-25	20000	1.16086349999999987

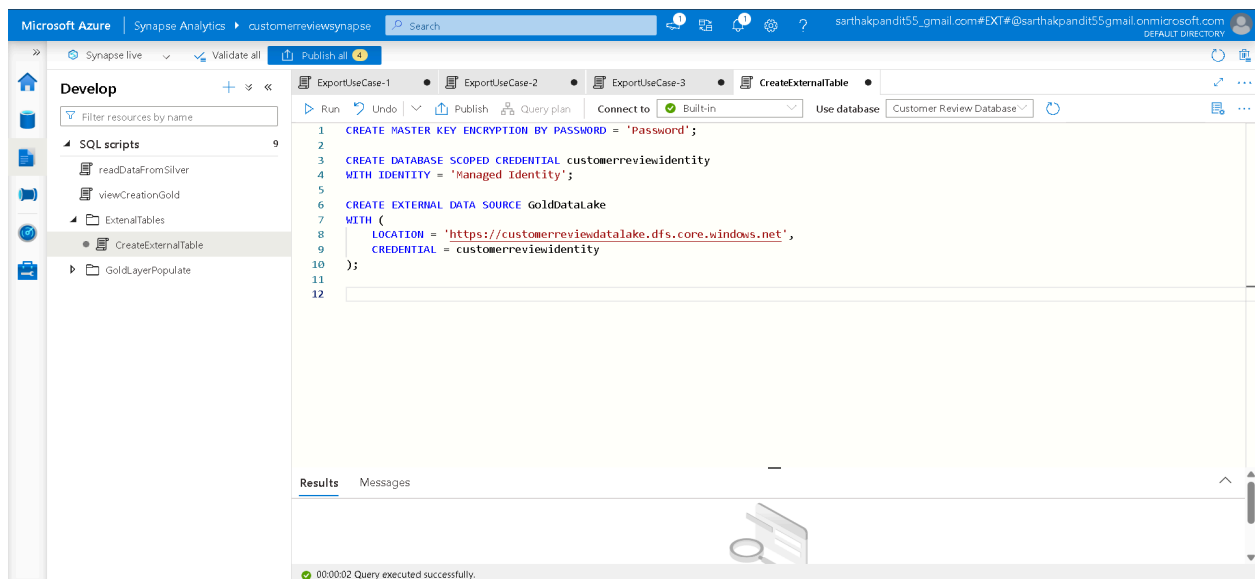
000001 Query executed successfully.

Task5 :- Creating External Tables and storing data in Datalake.

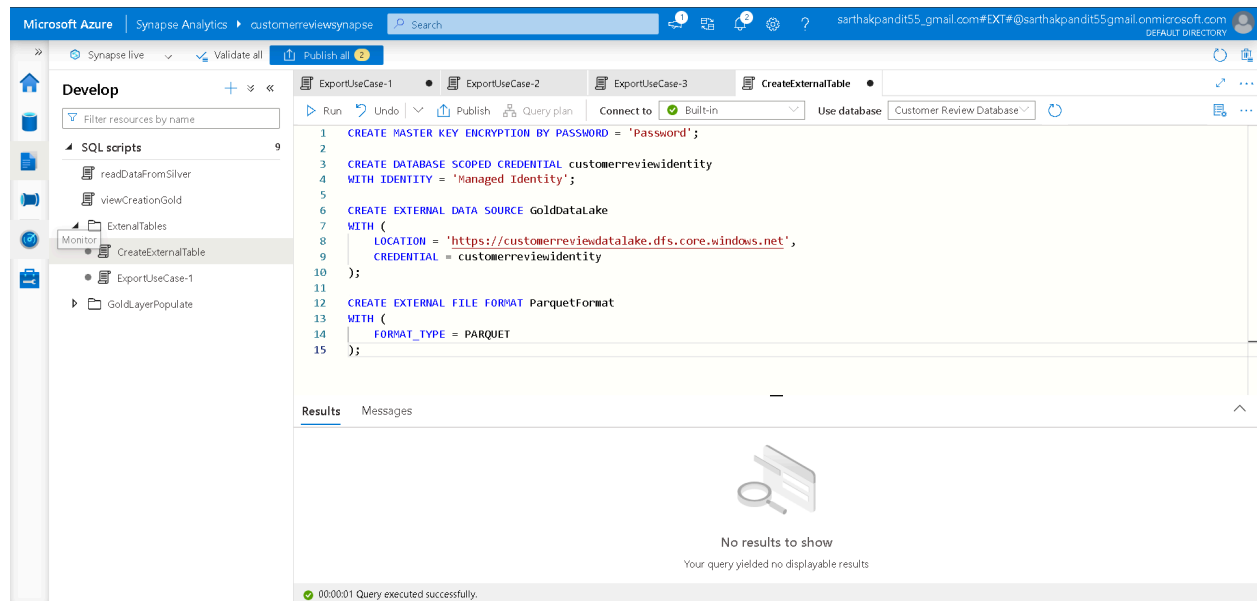
Step 1 :- Create a folder “**ExternalTables**”. In this folder create a file named “**CreateExternalTable**” and write the query and create “**Master Key**”.



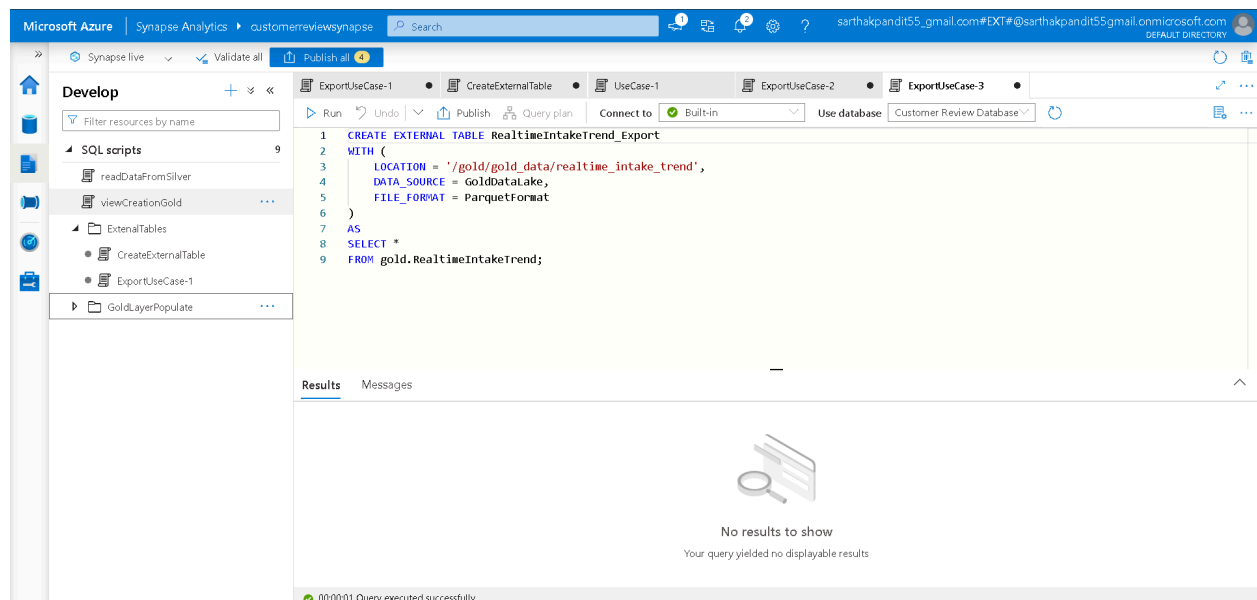
Step 2 :- Created Database Scope Credential and External data source.



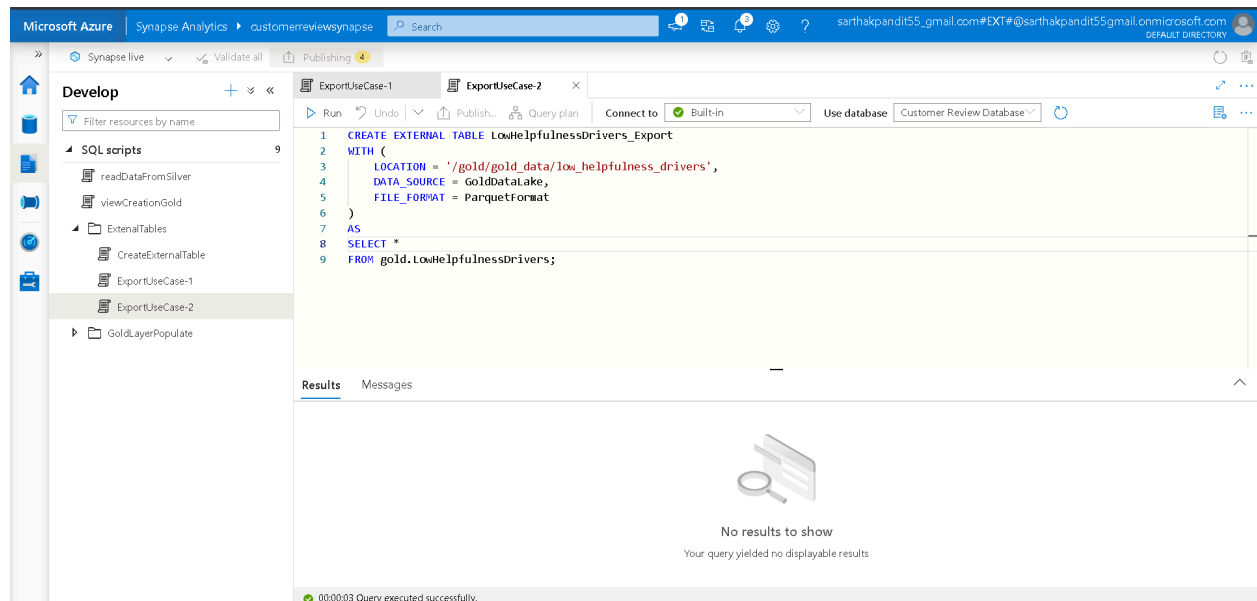
Step 3 :- Create “Parquet” file format as external tables do not support writing in “Delta” format.



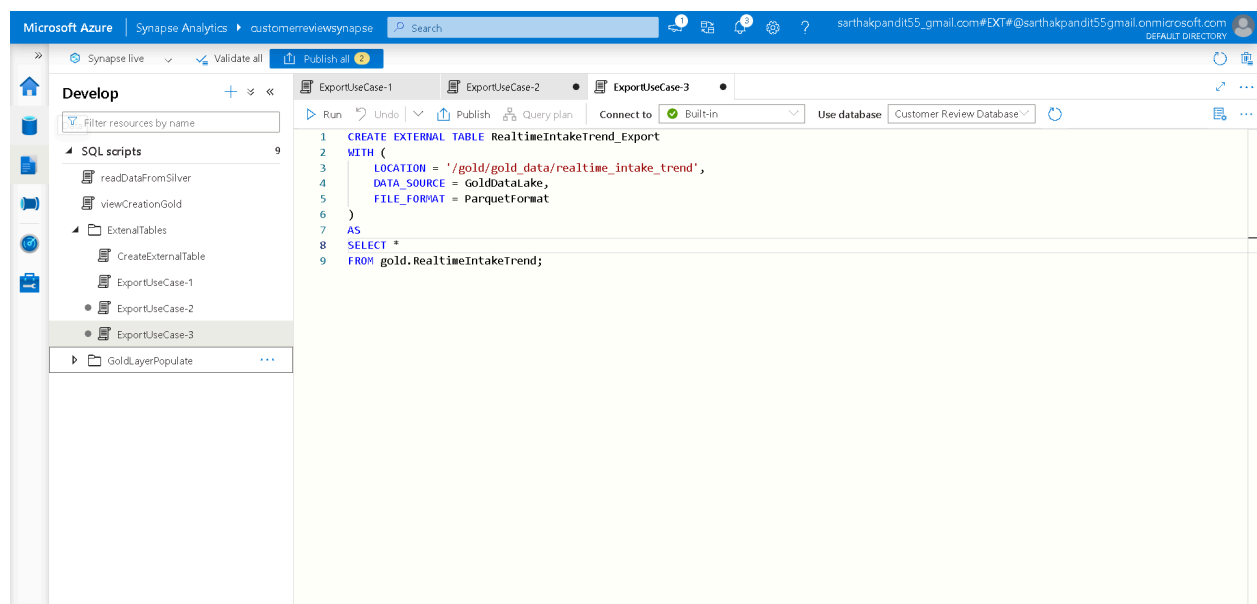
Step 4 :- Create “ExportUseCase-1” script file and write the script to create an external table and store data in “DataLake” .



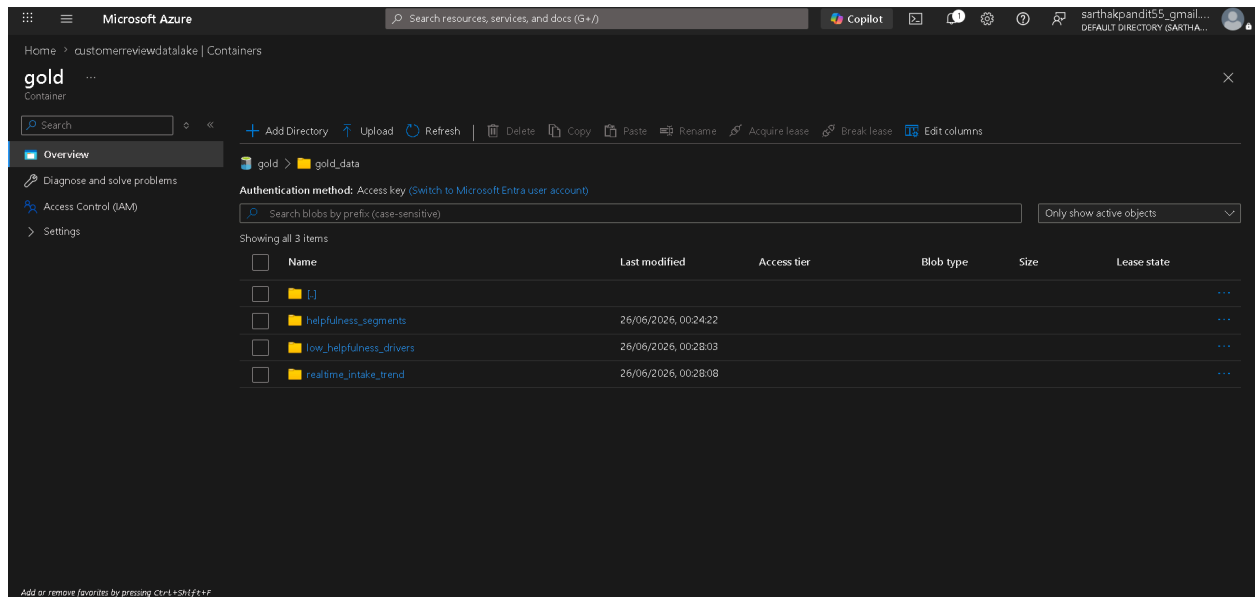
Step 5 :- Create “ExportUseCase-2” script file and write the script to create an external table and store data in “Datalake”.



Step 6 :- Create “ExportUseCase-3” script file and write the script to create an external table and store data in “Datalake”.



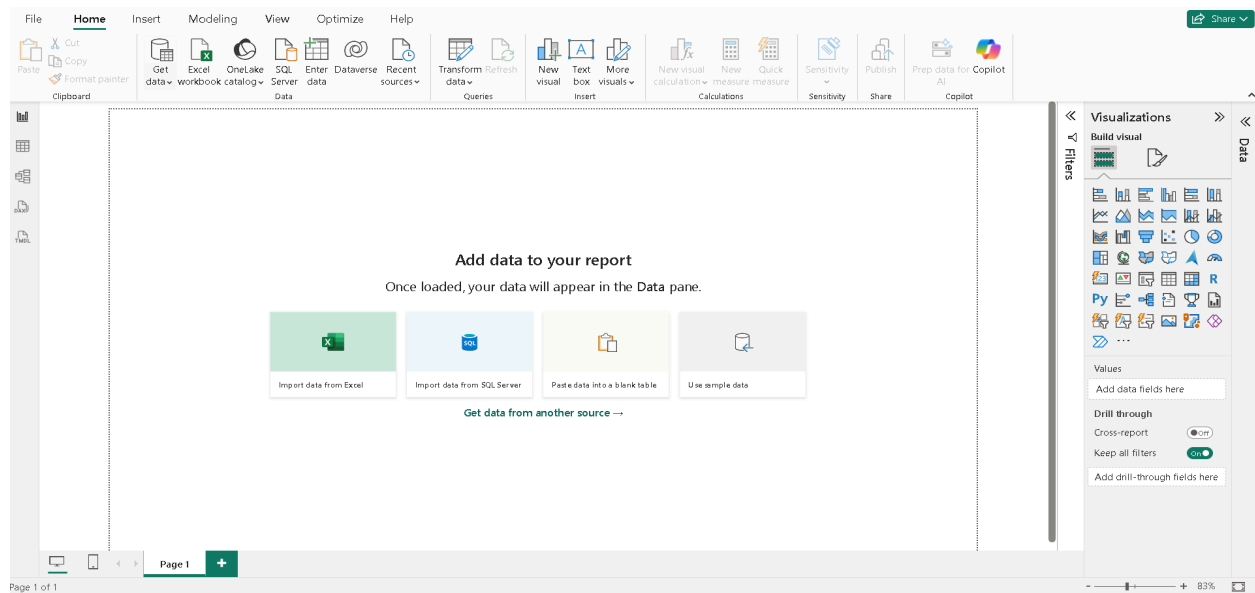
Step 7 :- All the data is stored in the gold container of “customerreviewdatalake” storage account in “Parquet” format.



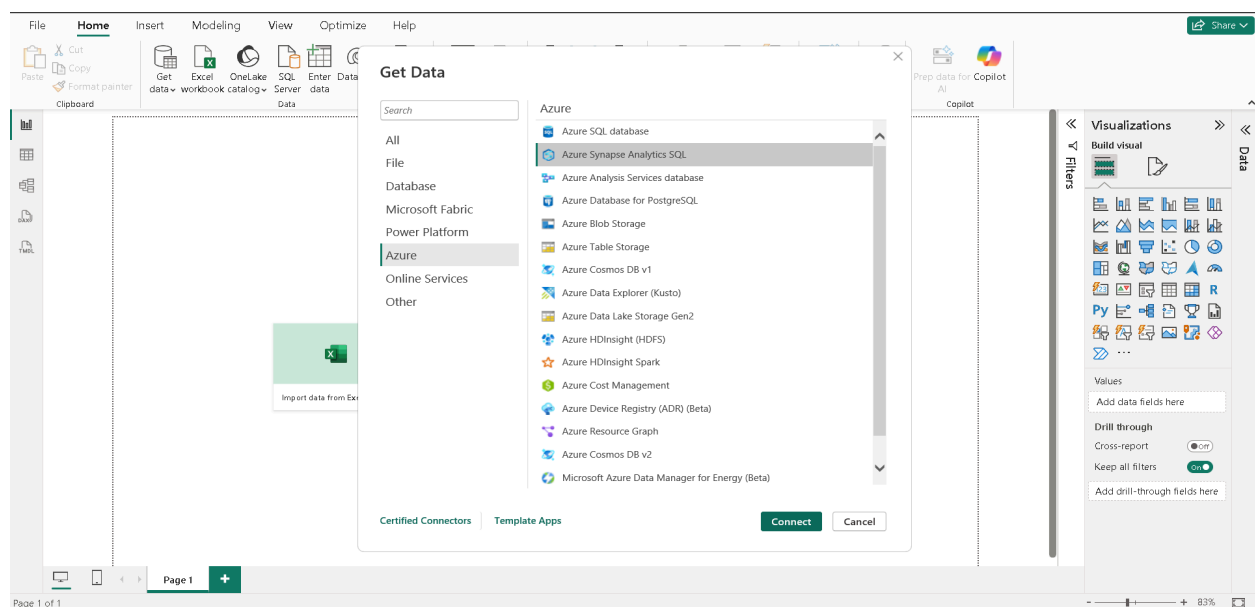
Power BI

Task1 :- Connecting Power BI with Synapse Analytics

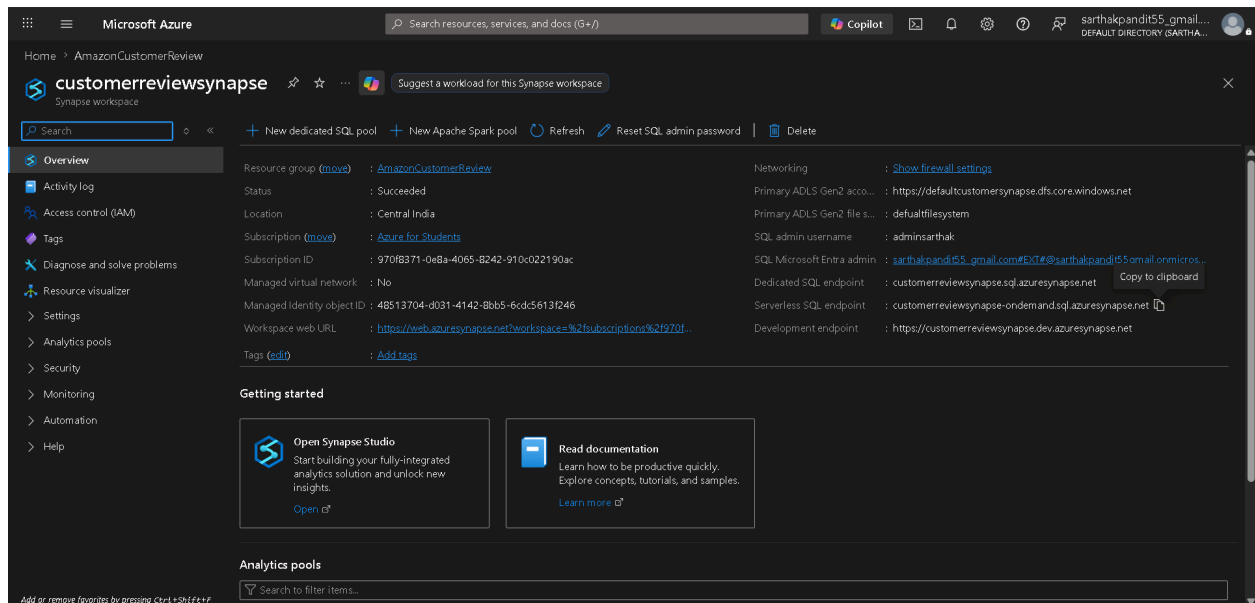
Step 1 :- Open PowerBI Desktop then click “**Get data**”.



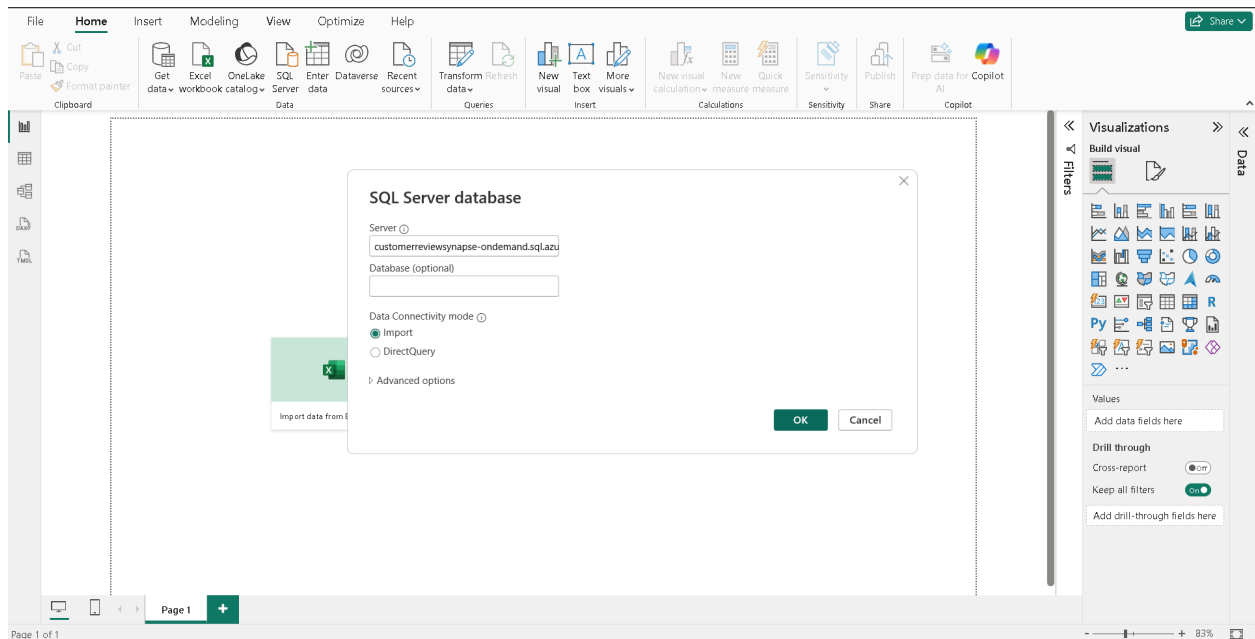
Step 2 :- Go to “**Azure**” tab and select “**Azure Synapse Analytics SQL**” and click “**Connect**”.



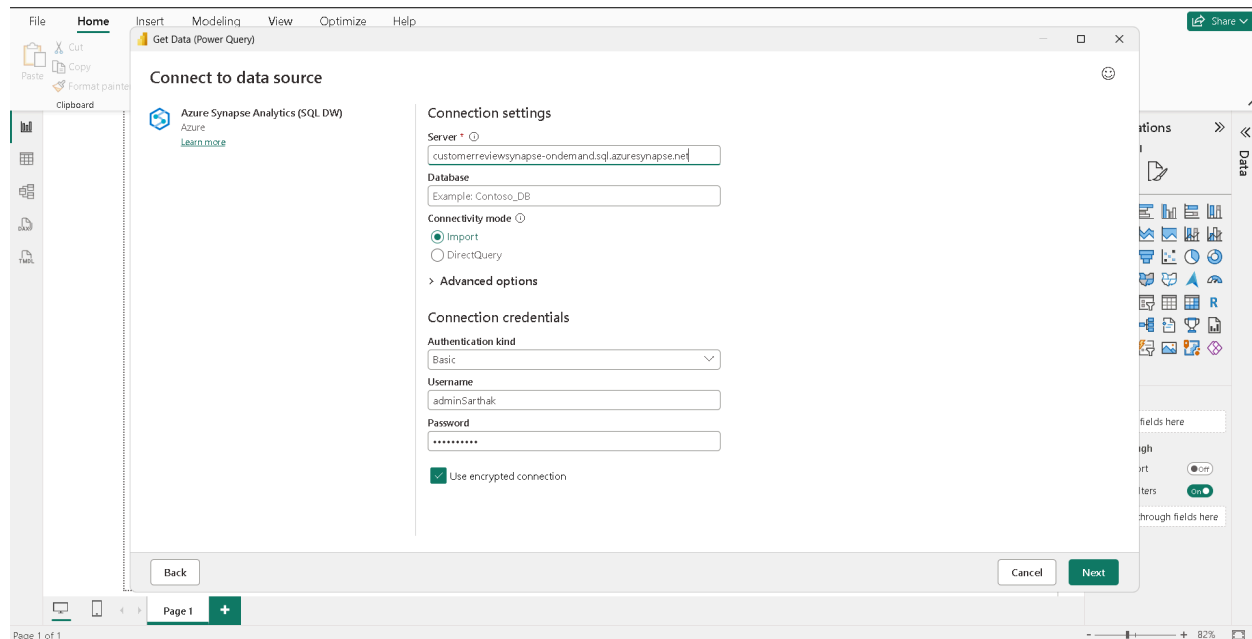
Step 3 :- Now go to “**Azure Synapse Analytics**” and copy “**Serverless SQL endpoint**”.



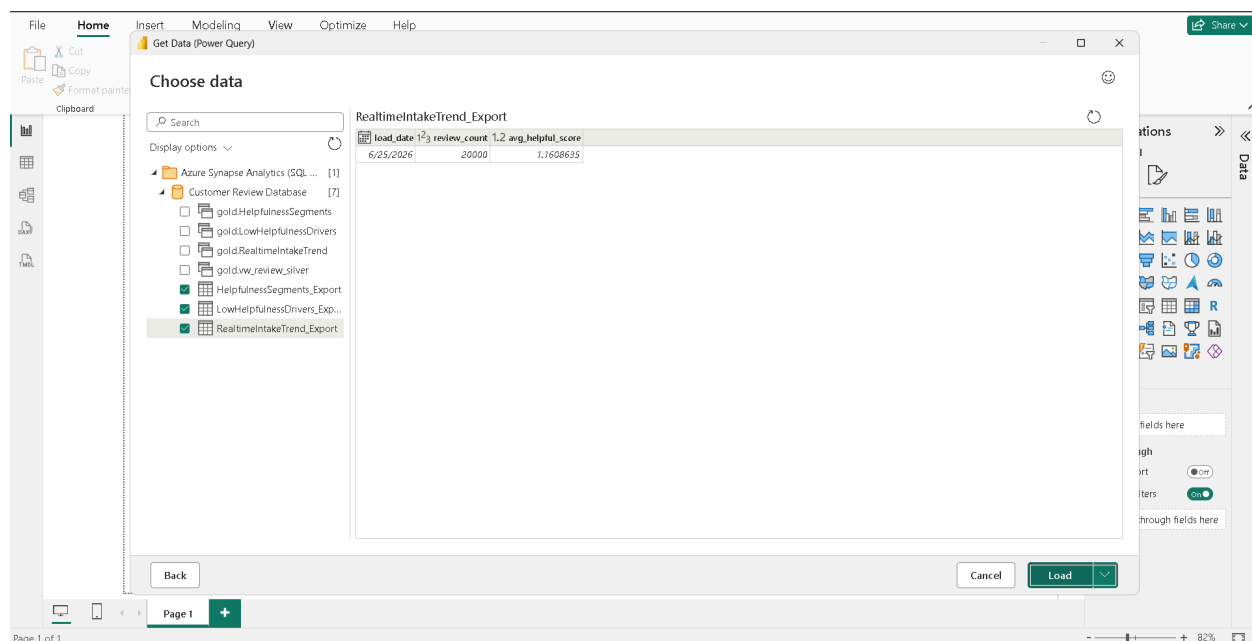
Step 4 :- Paste the “**Serverless SQL endpoint**” and click OK.



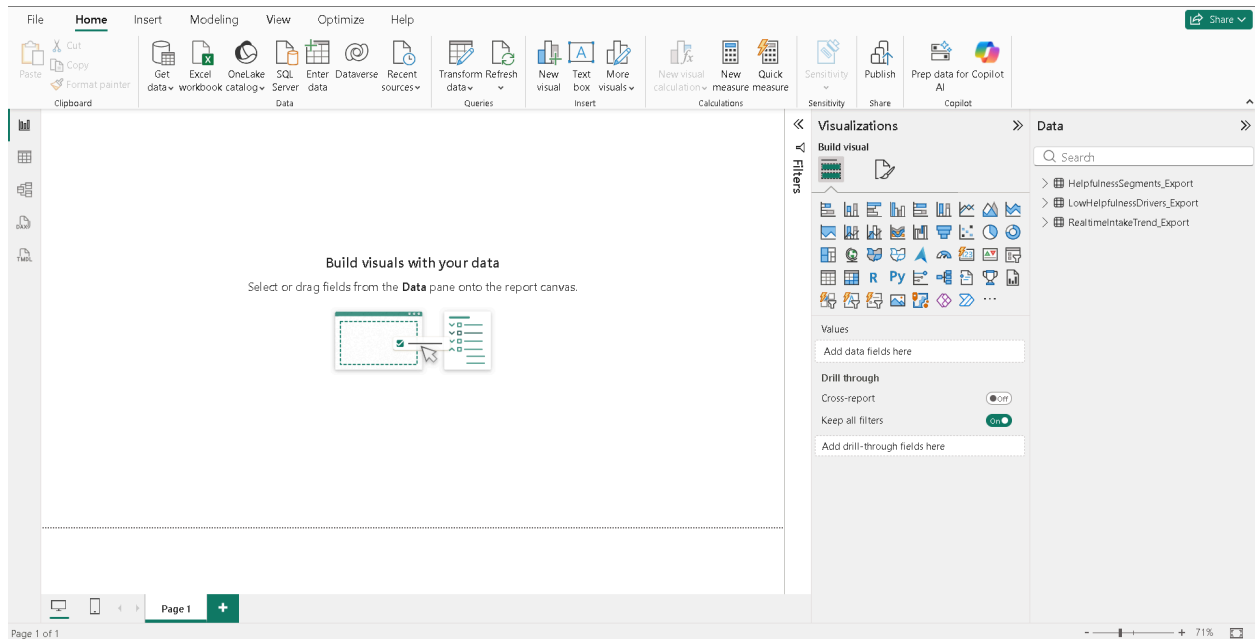
Step 5 :- Paste the “**Serverless SQL endpoint**” and enter the username and password of the database that is created while creating Synapse.



Step 6 :- After the successful connection with “**Synapse Analytics**”, Select the table to plot the graphs as per requirement and click “**Load**”.



Step 7 :- Tables are successfully loaded in Power BI Desktop.

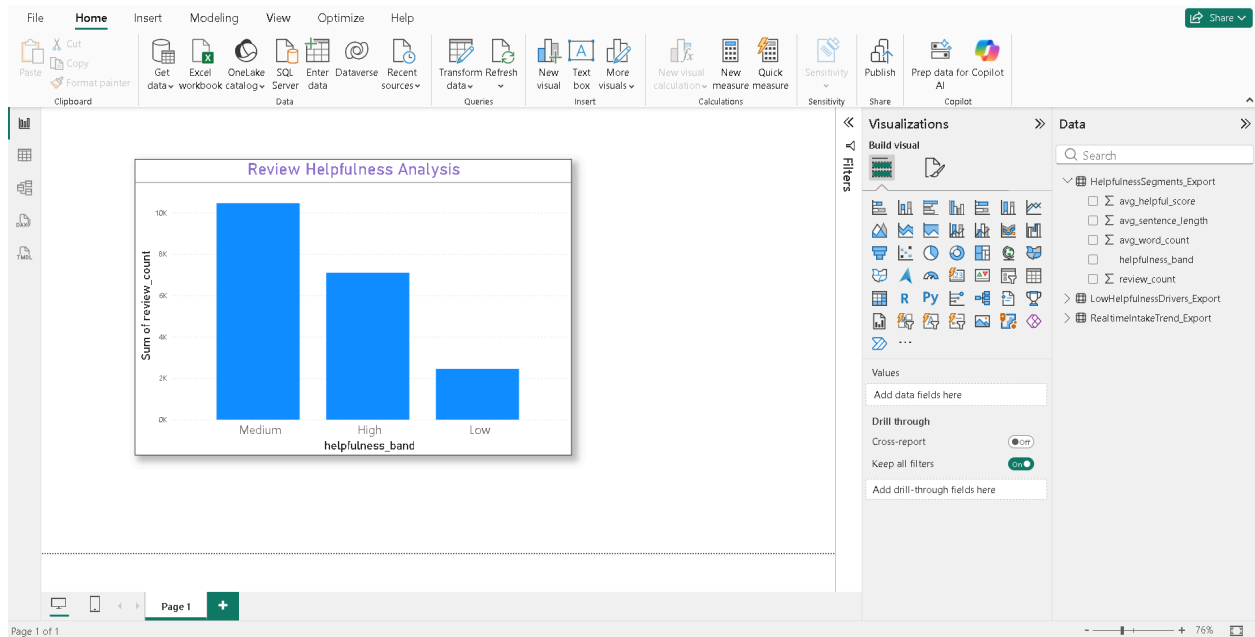


Task2 :- Business Logic Graphs

Graph 1 :- Review Distribution by Helpfulness

Business Insight :-

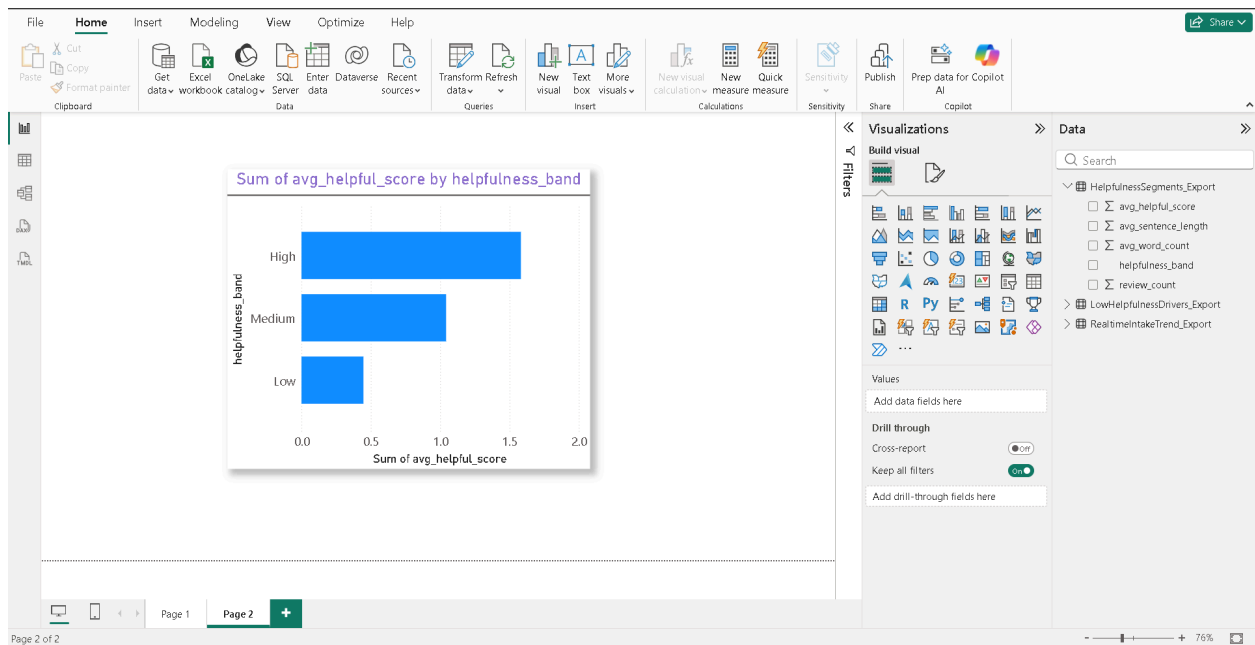
Helps identify whether customers are writing valuable reviews or whether review quality needs improvement.



Graph 2 :- Average Helpful Score by Segment

Business Insight :-

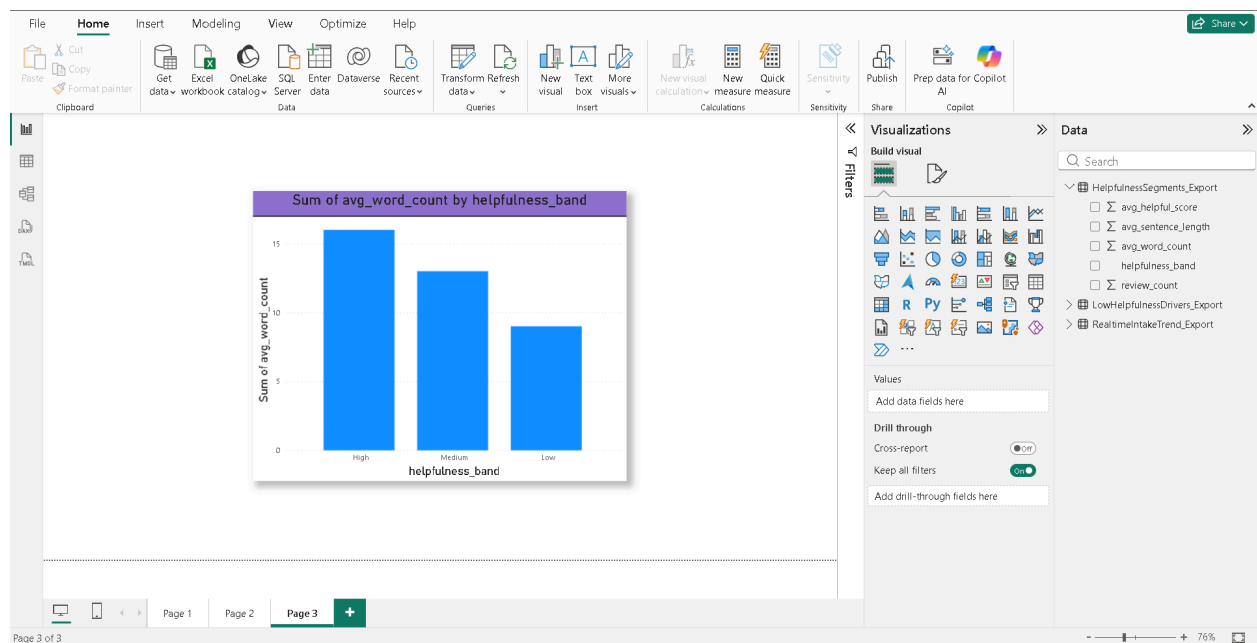
Highlights which category contributes the highest-quality reviews.



Graph 3 :-Average Review Length by Helpfulness.

Business Insight :-

Helps understand whether encouraging detailed reviews can improve review usefulness.



Graph 4 :- Low Helpfulness Drivers.

Business Insight :-

Identifies the biggest causes of poor-quality reviews, enabling targeted improvements such as prompting users to write longer, more descriptive feedback.

